

Sharing the Pain? Credit Supply and Real Effects of Bank Bail-ins

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We analyze the credit supply and real effects of bank bail-ins by exploiting the unexpected failure and subsequent resolution of a major Portuguese bank. Using loan-level data, we show that while firms more exposed to the bail-in suffered a significant contraction of credit at the intensive margin, they were on average able to compensate for the supply-driven shock. However, affected SMEs experienced a binding reduction of funds available through credit lines, and those with lower internal liquidity increased precautionary cash holdings and reduced investment and employment. Our results highlight the trade-off policymakers face when considering this new bank resolution mechanism. (*JEL* G01, G21, G28, G38)

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The recent financial crisis highlighted the pressing need for a robust mechanism to resolve banks in distress. In the absence of a viable alternative to insolvency that could lead to contagion and a credit crunch, policymakers around the world opted to bail out financial institutions using public funds. In Europe, for instance, taxpayers have covered more than two-thirds of the cost of recapitalizing and resolving banks (Philippon and Salord 2017). These interventions were often accompanied by significant government losses and austerity programs associated with political frictions and distributional problems. To counter this pervasive issue, most developed economies recently introduced formal resolution mechanisms with bank bail-ins that involve the participation of shareholders and creditors in bearing the costs of restoring distressed banks and include severe restrictions on taxpayer support.

An effective bank resolution framework should balance the benefits of imposing market discipline with the potential costs an intervention can have for credit supply and the real economy (Beck 2011).¹ In fact, Lambrecht and Tse (2019) show theoretically that, relative to bank bailouts, bail-ins reduce managerial risk-taking and improve loan quality but also lead to lower levels of credit provision and value created net of recapitalization costs.² This trade-off is explained by government guarantees that subsidize the cost of borrowing and artificially boost loan issuance and overall growth in a bailout regime. However, despite these theoretical predictions, there is little to no empirical evidence on the effects bank bail-ins might have on credit supply and economic activity. Our study contributes to filling this gap in the literature by analyzing the effects of the bail-in of a major bank in Portugal using a unique data set combining firm–bank matched data on credit exposures and interest rates with balance sheet information for firms and their lenders.

Specifically, we exploit the unexpected collapse of Banco Espírito Santo (BES) in August 2014—described as “one of Europe’s biggest financial failures” (FT 2014)—and subsequent resolution splitting the institution into a “good bank” and “bad bank.” While the intervention protected taxpayers and depositors, it left shareholders and junior bondholders holding the toxic assets and also affected to a lesser extent other resident banks that had to provide *ad hoc* funding to the Portuguese Resolution Fund. In line with the Bank Recovery and Resolution Directive (BRRD) of the European Union (E.U.), this bail-in was part of a strict and broad restructuring process that not only changed the bank’s ownership structure, but also automatically suspended

¹ Previous literature has shown the detrimental impact of public guarantees on bank risk-taking (e.g., Gropp, Hakenes, and Schnabel 2011; Dam and Koetter 2012; Silva 2019) as well as the negative effects of bank failures on real outcomes (e.g., Bernanke 1983; Calomiris and Mason 2003; Ashcraft 2005).

² Similarly, Berger et al. (2020) show that bail-ins provide superior capital incentives for banks relative to bailouts, while Schäfer, Schnabel, and Weder (2017) find evidence consistent with bail-in events reducing bailout expectations and thus increasing banks’ risk premiums.

the members of the bank's Board of Directors and Board of Auditors, with the new members of the management bodies appointed by the Portuguese Central Bank (Garcia 2016). Importantly, the bank's collapse was unrelated to fundamental risks in the Portuguese banking sector or in a generalized group of its borrowers. Instead, it was caused by large risky exposures to a limited number of foreign nonfinancial firms that were part of the Espírito Santo Group owned by the same family (Economist 2014). In fact, the Portuguese Central Bank indicated that the losses reflected a "fraudulent funding scheme between the companies belonging to the Group" not subject to its direct supervision and the "practice of management acts seriously detrimental to the interests of the bank and noncompliance with determinations prohibiting an increase of the exposure to other entities of the Group" (Banco de Portugal 2014a, 2014b). Thus, from an identification standpoint, this shock can be seen as exogenous from the perspective of any given Portuguese firm.

We start the analysis by examining over 115,000 bank-firm lending relationships and exploiting the widespread presence of firms borrowing from multiple banks, thus allowing us to control for changes in observable and unobservable firm characteristics such as credit demand, quality, and risk (Khwaja and Mian 2008). We find that firms more exposed to the bail-in suffered an economically and statistically significant supply-driven contraction of total committed credit and granted credit lines at the intensive margin.³ This effect is more pronounced among large firms and is concentrated on firms whose main lender prior to the shock was the bailed-in bank—that is, we find no evidence of a credit supply contraction for firms whose main lender was any other bank operating in the country, irrespective of whether the main bank participated in the *ad hoc* loan to the Portuguese Resolution Fund or not. Similarly, we find an economically large tightening of credit conditions after the bail-in, including higher interest rates on all new credit operations and granted credit lines, shortening of loan maturities, and an increase in the share of collateralized credit.

A fundamental follow-up question is whether more exposed firms could compensate for the credit supply tightening and worse credit conditions by accessing funds from other banks.⁴ Importantly, following Abowd, Kramarz, and Margolis (1999) and Cingano, Manaresi, and Sette (2016), we are still

³ Total committed credit between a bank and a firm refers to both drawn and undrawn credit, while granted credit lines include only the amount committed on revolving credit lines. In line with the existing literature, we consider granted instead of drawn credit throughout since the latter is likely driven by firm demand for credit—for instance, firms might actively choose to draw down credit lines, particularly in stress periods. Our conclusions do not change when using the complete sample of borrowing firms in Portugal in a model that replaces firm fixed effects with industry-location-size fixed effects for single-bank-relationship firms following Degryse et al. (2019).

⁴ This issue is particularly important in the context of small- and medium-sized enterprises (SMEs), which usually find it difficult to substitute credit from other sources because they are more opaque and thus mainly rely on existing banking relationships.

able to control for credit demand at the cross-sectional level by including in the regressions the vector of estimated firm-level fixed effects from the Khwaja and Mian (2008) within-firm specification. We find that, on average, firms more exposed to the bail-in did not suffer a reduction of overall credit after the intervention when compared to less exposed firms. In fact, the average firm was able to compensate for the reduction of total committed credit at the intensive margin with borrowing from other financial institutions with which it already had a relationship. However, when isolating granted credit lines from total credit and focusing on firms with credit lines at multiple banks, we show that SMEs more exposed to the resolution experienced a binding contraction in the quantity of funds available through lines of credit—an essential component for corporate liquidity management due to their unique role as insurance against future liquidity needs (Berg et al. 2017; Sufi 2009; Jiménez, Lopez, and Saurina 2009). Specifically, a one-standard deviation increase in firm exposure to the bail-in is associated with a 2.2% binding decrease in granted credit lines to SMEs. Our cross-sectional results also show that the resolution came at the cost of modestly worse credit conditions for more exposed firms.

Finally, we also find evidence of a negative adjustment of investment and employment policies at SMEs more exposed to the bail-in. This effect is again economically significant, with a one-standard deviation increase in firm exposure to the shock leading to a relative drop in investment and employment at SMEs of up to 2.0% and 1.5%, respectively. These adverse effects of the bank resolution on real sector outcomes are driven by a response to increased liquidity risk by firms with lower *ex ante* internal liquidity. Consistent with the argument that firms' option to access credit lines should be more valuable when internal liquidity is scarce (e.g., Campello et al. 2011), we find that the negative real effects are concentrated among less liquid SMEs more exposed to the resolution that responded to the shock by increasing cash holdings while decreasing investment and employment. Instead, in line with precautionary cash savings being important in times of dislocation in markets for external finance (e.g., Duchin, Ozbas, and Sensoy 2010), more exposed SMEs with high liquidity before the bail-in were able to use their available internal cash holdings to compensate for the binding contraction in granted credit lines and thus maintain steady levels of employment and investment.⁵

The contribution of this paper is twofold. First, we contribute to the financial intermediation literature by examining how distressed banks should be resolved. Kahn and Winton (2004) suggest that a “good-bank-bad-bank”

⁵ In a separate but related exercise, we gauge whether the bailout of four Portuguese banks in 2012 resulted in similar negative effects. We find no significant differences between borrowers of bailed-out and non-bailed-out banks in terms of credit supply, investment, or employment. This points to rather sharp differences between the two bank resolution policies, although we caution that (i) the macroeconomic situation was considerably different during these two episodes; (ii) in contrast with the 2012 bailouts, the bail-in in 2014 was accompanied by management and governance changes; and (iii) the public intervention in 2012 was more systemic in nature.

split may be beneficial, reducing risk-shifting incentives in the healthy bank and increasing its incentive to screen and monitor the performing loans. More recent work, however, has mostly focused on describing the potential benefits and costs of the different bank resolution mechanisms (e.g., Dewatripont 2014; Conlon and Cotter 2014; Avgouleas and Goodhart 2015; Philippon and Salord 2017; Beck, Radev, and Schnabel 2020) and examining the interaction between bail-ins and bailouts from a theoretical perspective (e.g., Berger et al. 2020; Bernard, Capponi, and Stiglitz 2019; Colliard and Gromb 2019; Keister and Mitkov 2019; Klimek et al. 2015; Lambrecht and Tse 2019; Segura and Vicente 2019; Walther and White forthcoming). Our paper contributes to this literature by empirically assessing the effects of a bank bail-in on credit supply and real sector outcomes using detailed bank-, firm-, and loan-level data.

Second, our paper contributes to the corporate finance literature on firms' liquidity management, highlighting the "dark side" of precautionary savings in the context of a supply-driven credit line reduction as well as the importance of firms' internal liquidity for the transmission of credit supply shocks to the real economy. Under the precautionary demand for cash theory, firms hold cash as a buffer against adverse cash flow shocks. This can be particularly valuable for firms that are financially constrained (Almeida, Campello, and Weisbach 2004) and following a credit crunch (Duchin, Ozbas, and Sensoy 2010). However, Berg (2018) finds that while liquid SMEs are indeed able to absorb credit supply shocks by using existing cash buffers, their illiquid counterparts increase cash holdings when a loan application is rejected, cutting noncash assets by more than the requested loan amount, and thus investment and employment. While Berg (2018) uses discontinuities in credit scores comparing accepted and rejected loan applicants at a single German bank, we generalize his findings by using an exogenous bank shock for identification and analyzing the entire banking sector of an economy.

1. Background

1.1 Bank failure and subsequent resolution

Banco Espírito Santo (BES) collapsed and was put into resolution by the Portuguese Central Bank on August 3, 2014. This decision was the result of a rapid series of events including the disclosure of hefty losses of €3.6 billion in the first half of 2014 arising from exposures to Espírito Santo Financial Group (ESFG)—the parent holding company set up outside Portugal to hold investments of the Espírito Santo family, including a 20% stake in the bank.⁶

⁶ While the stake of the family at the bank was held by Luxembourg-based ESFG, this company was in turn owned by Espírito Santo International (ESI)—the Espírito Santo family's ultimate holding entity, also headquartered in Luxembourg. Intertwined into this corporate structure was ES Bank Panama, a subsidiary of ESFG that extended hidden credit lines to ESI-connected companies (FT 2014).

At the time of the resolution, the bank was the third-largest financial institution in Portugal, with a market share of 19% of credit granted to nonfinancial corporations and 12% of total deposits (Banco de Portugal 2014b).⁷ It was also one of only four financial intermediaries in Portugal classified as a significant credit institution by the European Central Bank (ECB).⁸

The scale of the losses came as a surprise to the Portuguese Central Bank, which indicated that these reflected a “fraudulent funding scheme between the companies belonging to the Group” not subject to its direct supervision and the “practice of management acts seriously detrimental to the interests of the bank and noncompliance with determinations prohibiting an increase of the exposure to other entities of the Group” (Banco de Portugal 2014a, 2014b).⁹ The losses placed the bank’s regulatory capital ratios considerably below those required by the Portuguese Central Bank and the European Banking Authority (EBA), and the ECB’s decision to suspend the failing bank as a counterparty in monetary policy operations made its liquidity shortfall situation unsustainable.

The resolution included the immediate creation of a bridge bank named Novo Banco that received sound assets and liabilities such as cash, retail deposits, performing loans, and central bank funding. In contrast, shareholders and junior bondholders were bailed in and thus left with the toxic assets that led to the mounting losses, which remained in a “bad bank” that was liquidated. Importantly, the newly created bank became fully owned by the Portuguese Resolution Fund, which provided the entirety of the €4.9 billion of capital. The financial resources of the Fund do not include public money, as it is financed by the initial and periodic contributions of all of the country’s lenders as well as the proceeds from a levy on the banking sector (Banco de Portugal 2014a). However, since the Fund was only established in 2012, it did not yet have enough resources to fully finance such a large operation and, as a result, obtained a loan from a syndicate of eight member banks (€0.7 billion) and another from the Portuguese government (€3.9 billion). Both loans were made to the Portuguese Resolution Fund and not to the bank itself, with the country’s lenders who bear

⁷ As a universal bank, both before and after the resolution, the institution provides a diversified range of financial services such as domestic commercial banking to retail, corporate, and institutional clients, as well as private banking, international commercial banking, asset management, and insurance.

⁸ The major objectives of the Portuguese authorities when designing this intervention were therefore to (i) safeguard the interests of taxpayers and ensure accountability of the bank’s shareholders; (ii) protect depositors, for whom, unlike in Cyprus (e.g., Brown, Evangelou, and Stix 2018), potential losses were completely ruled out; and (iii) preserve financial stability (Banco de Portugal 2014b).

⁹ As highlighted by the Governor of the Portuguese Central Bank, these losses largely exceeded the foreseeable values in light of the information disclosed until then by the bank and its external auditor (Banco de Portugal 2014a). In fact, only two weeks before disclosing the substantial losses, the bank announced that its exposure to the family-controlled group of nonfinancial entities amounted to just €1.2 billion—based on this information, the bank’s capital buffer at the time (€2.1 billion) would have been sufficient to accommodate any possible negative shocks arising from such exposure.



Figure 1
Evolution of bank CDS spreads over time

This figure plots daily five-year CDS spreads on senior unsecured debt between 2010:Q1 and 2015:Q4. “Bailed-in bank” refers to BES until August 3, 2014 (i.e., the day of the resolution—vertical dashed line), and the “good bank” (Novo Banco) thereafter. CDS spreads for “Other Portuguese banks” are computed as the equal-weighted average across banks in Portugal with available information (Caixa Geral de Depósitos, Banco BPI, Banco Millennium BCP). The banks in this figure correspond to the four significant credit institutions operating in Portugal as defined by the ECB. Source: IHS Markit.

the risks of the operation having several years to recoup the shortfall with their ordinary contributions (Banco de Portugal 2014a; Garcia 2016).¹⁰

1.2 Unexpected and idiosyncratic nature of the bank failure

Figure 1 shows that the credit default swap (CDS) spreads of the bailed-in bank moved in line with the rest of the banking sector until late June 2014, when the degree of exposures to the holding company owned by the same family started to be revealed. Within a month, the spreads moved from less than 2% to almost 7%. The event came after a long period of increasing stability in the

¹⁰ The Portuguese Central Bank decided to move even further toward a bail-in type of intervention with a re-resolution in the last days of 2015—16 months after the original intervention. Specifically, a limited number of bonds were transferred to the “bad bank,” imposing losses on almost €2 billion of senior bondholders. In October 2017, Lone Star Funds (a US private equity fund) acquired 75% of the “good bank” in return for a capital injection of €1 billion, with the remaining 25% held by the Portuguese Resolution Fund. Given that we only have loan- and firm-level data available through 2015, our analysis does not consider these two shocks and is instead solely focused on the effects of the original resolution in August 2014.

banking sector, with CDS spreads for Portuguese banks having declined from their crisis peak of around 16% in late 2011.¹¹

The figure also shows the limited contagion effects of the resolution to the remainder of the banking system, with the average CDS spread of the other local banks considered significant credit institutions by the ECB increasing only slightly in the weeks leading up to the intervention and remaining below 3.5% until the end of 2015. This pattern is consistent with Hüser et al. (2018) that, using granular data on the securities cross-holdings among the largest euro area banking groups, show that bail-ins lead to limited spillovers due to low levels of securities cross-holdings in the interbank network and no direct contagion to creditor banks. Similarly, Havemann (2019) shows that the bail-in of a small South African lender financed almost exclusively with wholesale funding had a limited impact on the financial system in terms of contagion. Nevertheless, to be conservative in our analysis, we take into account the exposure of other banks to the bail-in—even if small—through the institution-specific amount of financing to the *ad hoc* loan granted to the Portuguese Resolution Fund.

1.3 Bail-in versus losses prior to the resolution

A key concern is whether our estimates are driven by the bail-in per se or, instead, by the losses at the bailed-in bank that occurred prior to the resolution. In fact, there is a large literature in different settings showing that banks cut lending after a shock to their net worth, which then leads to negative real effects (e.g., Peek and Rosengren 2000; Puri, Rocholl, and Steffen 2011; Chodorow-Reich 2014; Huber 2018). In our case, however, the bail-in should indeed be the driver of the results because of the division of assets into a “good bank” and a “bad bank.” Specifically, the shock we study is rather unique since the €3.6 billion of losses and related bad assets associated with risky lending to a few nonfinancial foreign entities of the Espírito Santo Group were fully absorbed by the “bad bank” that was liquidated. In other words, the toxic exposures and respective losses were not passed to the balance sheet of the “good bank” that started operating after the resolution. Thus, we should not expect our results to be driven by the bailed-in bank changing its lending policies after the resolution directly as a result of the losses disclosed a few weeks prior to the intervention.¹²

¹¹ At the country level, by the end of the EC/ECB/IMF Economic Adjustment Program in June 2014, Portugal was growing 0.3% faster than the European Union, excluding Germany (Reis 2015).

¹² Empirically, the exposures to nonfinancial entities part of the Espírito Santo Group that generated the hefty losses and drove the collapse of the bank are not part of our estimations since (i) these firms were based abroad and our data set only captures credit granted to firms headquartered in Portugal and (ii) given that their credit claims were fully transferred to the “bad bank” that was liquidated, these would not appear in the postshock period of our difference-in-differences specification even if a firm was based in Portugal. Note that the other major shareholders and junior bondholders of the bank wiped out during the resolution were also foreign institutions. Besides ESFG—the family-controlled holding company based in Luxembourg and the largest stockholder with a 20% stake—the other main shareholders of the bank included Credit Agricole (France) with 15%, Silchester International Investors (U.K.) and BlackRock (U.S.) with 5% each, and Capital Research and Management Company (U.S.) and Bradport (fully owned by Banco Bradesco based in Brazil) with 4% each (BES 2014).

Similarly, the “good bank” was properly recapitalized during the resolution and had a Tier 1 capital ratio of 10.3% when it started operating—above the regulatory minimum and in line with capital ratios of other major banks in Portugal and in Europe at the time. In addition, the bailed-in bank was also not capitalized worse than its preshock counterpart, with BES having a Tier 1 capital ratio of 10.5% in 2012:Q4 and 10.6% in 2013:Q4. Unlike in the literature exploiting shocks to bank capital requirements (e.g., Aiyar, Calomiris, and Wieladek 2014; Jiménez et al. 2017; Gropp et al. 2019), there were also no changes in the regulatory framework that affected the bailed-in bank differently than other banks in Portugal during the period we analyze. Thus, given the evidence on the effects of different bank interventions (e.g., Giannetti and Simonov 2013; Augusto and Félix 2014) and that the capital injection was large enough to re-establish bank capital requirements, we should also not expect our results to be driven by the bailed-in bank changing its lending policies after the resolution due to a lack of sufficient capital.

1.4 Bail-in versus bailout

The bank bail-in we analyze differs markedly from the bailouts of distressed banks funded by taxpayers around the world, including in Portugal, following the 2007–2009 financial crisis. Given that the resolution imposed the losses on shareholders and creditors, this holds even when being overly conservative and considering this intervention a potential hybrid of a bail-in and bailout (Economist 2014).

In addition, a bail-in is part of a stricter and broader restructuring process that not only wipes out shareholders and creditors—thus changing a bank’s ownership structure—but also replaces managers, thus altering the bank’s governance structure. The fact that the resolution authority in a bail-in has the legal power to replace the failing bank’s management is key since it changes substantially the risk-taking incentives at the bank. The new resolution framework also requires the newly appointed managers to engage in an extensive reorganization plan to address the bank’s structural weaknesses and restore its long-term viability. The basis for the reorganization strategy should be the factors that caused the entry of the bank into resolution and include withdrawal from loss-making activities and stronger risk-management framework, capital, and liquidity planning.

Overall, we should expect a distressed bank to be considerably more prudent moving forward if it is resolved through a bail-in rather than a government bailout. While in theory a bailout can also come with significant changes to the bank’s governance structure (e.g., new managers, more emphasis on risk management and control), in practice this approach has only been prevalent for bail-ins where such management and governance changes are not an option but part of a statutory power in the new resolution regime that mandates such

changes to occur.¹³ This trend has been pervasive not only in Portugal,¹⁴ but also outside the country.¹⁵

In fact, free of the riskiest exposures and associated losses that were passed to the “bad bank,” the bailed-in bank broke with the former legacy by implementing a more sustainable lending policy.¹⁶ This included the reduction and nonrenewal of credit lines as well as a decrease in its credit exposure to large firms, with any new large loan requiring extensive review and risk committee board approval (Novo Banco 2014). The latter is particularly important since, as Berg (2015) shows, the involvement of risk managers in the loan origination process can reduce default rates by more than 50%. The bailed-in bank’s strategy to manage its capital and liquidity positions prudently by deleveraging noncore assets from the loan portfolio and optimizing risk-weighted assets led to an 18% reduction of its balance sheet from the day of the resolution until the end of 2015 (Novo Banco 2015).

Importantly, while this intervention occurred before transposition of the BRRD into national legislation, the Portuguese resolution regime then in force was already, in substance, very similar to the final European regulation (Garcia 2016). Unlike in a bailout, this included the automatic suspension of the members of the bank’s Board of Directors and the Board of Auditors, with the new members of the management bodies appointed by the Portuguese Central Bank, which also approved the statutes, strategy, and risk profile of the newly created institution.

Overall, this shock, widely considered one of the most prominent cases of a bank bail-in to date worldwide (e.g., Dell’Ariccia et al. 2018; Garcia 2016; Philippon and Salord 2017), arguably provides a unique laboratory to shed light on the potential effects of future similar resolutions.

¹³ The E.U. and the U.S. strengthened their bank resolution regimes and introduced bail-in powers via the BRRD and the Dodd-Frank Act, respectively. Despite many similarities between the two resolution schemes (e.g., removal of the management team of the failing financial institution), there are still some important differences such the lack of a restructuring option in the United States (Philippon and Salord 2017).

¹⁴ Unlike in the bail-in of BES in 2014, there were no management changes following the bailouts of four Portuguese banks in 2012. Specifically, the same individuals held the top management positions before and at least three years after the bailouts of Caixa Geral de Depósitos (CGD), Banco Millennium BCP, Banco BPI, and Banco Internacional do Funchal (BANIF). The only exception was CGD (the state-owned and largest bank in Portugal), with the Chairman of the Board of Directors voluntarily resigning and ceasing functions one year after the June 2012 bailout for reasons unrelated to the public intervention—in fact, the Chairman of Executive Committee remained the same.

¹⁵ In addition to the bail-in in Portugal, management and governance changes also occurred following the bail-in of (i) SNS Reaal, a Dutch bank nationalized in 2013 with claims of shareholders and certain junior bondholders wiped out; (ii) the small Danish cooperative bank Andelskassen in 2015; and (iii) the Spanish Banco Popular that was bailed-in and immediately sold to Santander for €1 in 2017, in the first resolution in Europe directly arranged by the EU’s Single Resolution Board (SRB). Bailouts, on the other hand, do not necessarily imply an immediate change in governance and management. Besides the bailouts of four Portuguese banks in 2012, other notable examples include the 2008 bailouts of Commerzbank in Germany (the CEO only stepped down in 2016) and Lloyds in the U.K. (the new CEO only entered into functions in 2011).

¹⁶ The detailed list of assets, liabilities, and off-balance-sheet items transferred to the “bad bank” beyond the exposures to the family-controlled parent group is available in Annex 2 of Banco de Portugal (2014b).

2. Identification Strategy

We investigate the credit supply and real effects of a bank bail-in in two steps. First, we assess whether the resolution induced significant changes in the supply of credit to firms at the intensive margin (within-firm analysis). Second, we investigate whether more affected firms were able to substitute funding from other banks, if they were able to obtain similar credit conditions, and the consequences of the intervention for investment and employment (cross-sectional analysis). The first part of the analysis uses firm–bank matched data to exploit variation within firms that have more than one lending relationship, while the second makes use of variation across firms with different preshock exposures to the bail-in.

2.1 Within-firm analysis

The main challenge of our empirical analysis is to identify the causal impact of bail-ins on credit supply, price conditions, and real outcomes. In fact, an intervention may be correlated with underlying changes in the overall economic situation that may affect both credit provision and firms' demand for credit. To address this identification problem, we exploit an exogenous shock in August 2014 corresponding to an unexpected bank failure and subsequent resolution and, following the novel approach of Khwaja and Mian (2008), use matched firm–bank loan-level data to account for unobserved heterogeneity in firms' credit demand, quality, and risk by saturating our model with firm fixed effects. As a result, our identification comes entirely from firms that were borrowing from at least two different banks before and after the resolution. This strategy isolates the causal impact of the bail-in shock on the change in credit supply by comparing the within-firm variation in the change in lending from banks differently exposed to the intervention. The baseline specification is defined as:

$$\Delta \log(Y)_{bi} = \beta(\text{BankExposure}_b) + \delta' X_b + \alpha_i + \varepsilon_{bi}, \quad (1)$$

where the main dependent variable $\Delta \log(Y)_{bi}$ is either the log change in total committed credit or the log change in granted credit lines from bank b to firm i from the pre- to the postshock period. In the latter case, our identification comes from the subset of firms (35%) with credit lines granted by at least two different banks before and after the resolution. We also use the change in interest rates, maturity, and share of collateralized credit as alternative outcome variables. As in Khwaja and Mian (2008), the data are collapsed (time-averaged) into a single pre- (2013:Q4–2014:Q2) and postresolution period (2014:Q3–2015:Q3) to ensure our standard errors are robust to auto-correlation (Bertrand, Duflo, and Mullainathan 2004).

The main independent variable, BankExposure_b , is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Portuguese Resolution Fund granted as part of the

resolution for the eight participating banks (as a percentage of assets), and zero otherwise. This indicator is computed using the publicly available financial statements of the banks operating in Portugal. The percentage of assets that was effectively bailed-in for the resolved bank amounts to 6.79%, while the specific contribution to the *ad hoc* loan to the Fund granted as part of the resolution for the eight participating banks ranges from 0.04% to 0.37% of assets—see Table IA1 in the Internet Appendix for the explicit value for each bank where this indicator exceeds zero.¹⁷ We do not include in this measure the ordinary contributions to the Fund that each bank made in 2013, as these were already priced in before the resolution. As in Silva (2019), for instance, we scale all coefficients throughout the paper by the corresponding variable's standard deviation to ease the interpretation of the magnitudes and ensure comparability across the different samples.

X_b refers to a set of bank-level controls measured in the pre period, including size (log of total assets), ROA (return-on-assets), capital ratio (equity to total assets), liquidity ratio (liquid to total assets), and NPLs (nonperforming loans to total gross loans). These controls are particularly relevant in our setting since bank-specific exposures to the bail-in are not randomly assigned but a function of bank characteristics (e.g., the contribution to the Fund is determined by each bank's amount of liabilities), which may be correlated with changes in their willingness to lend.¹⁸ α_i are firm fixed effects that capture firm-specific determinants of credit flows and can be interpreted as a measure of credit demand (e.g., Cingano, Manaresi, and Sette 2016). Finally, since the shock is bank-specific, changes in the credit granted from the same bank may be correlated. As a result, we use standard errors clustered at the bank level.

2.2 Cross-sectional analysis

Although the above specification allows us to examine whether there was indeed a credit contraction and which type of firms were more likely to be affected by the shock, it is not appropriate to assess the overall effect of the shock. This is because the within-firm analysis is not able to capture credit flows from new lending relationships and ignores all terminated lending relationships between the pre- and postshock period. Given the importance of the extensive margin for credit adjustment, we also estimate the related between-firm (cross-sectional) effect of firm exposure to the shock as:

$$\Delta \log(Y)_i = \beta(\text{FirmExposure}_i) + \tau' F_i + \delta' \bar{X}_i + \hat{\alpha}_i + \varepsilon_i, \quad (2)$$

where $\Delta \log(Y)_i$ is the log change in total committed credit or in granted credit lines between 2013:Q4 and 2015:Q4 from all banks to firm i . We use the same

¹⁷ For confidentiality reasons, we cannot directly compare one bank with all other banks.

¹⁸ While the largest banks generally provided more funds to the Portuguese Resolution Fund in absolute terms, there is little correlation with size when scaling the individual contributions with each bank's total assets—see Table IA1 in the Internet Appendix.

model to examine the effects of the bail-in on other credit conditions and the real economy.

$FirmExposure_i$ is the exposure of each firm to the bail-in, computed as the weighted average of $BankExposure_b$ across all banks lending to a firm, using as weights the preperiod share of total credit of each bank. F_i are firm characteristics including size (log of total assets), age ($\ln(1+age)$), ROA (net income to total assets), capital (equity to total assets), and liquidity (current assets to current liabilities)—all measured as of 2013:Q4. We also include industry and district fixed effects in the model. Bank controls $\bar{X}_i$ include the same variables as in specification 1 but are averaged at the firm level according to the share of total credit granted to the firm by each bank prior to the shock.

Finally, given that in the cross-sectional model 2 the firm-specific demand shock α_i cannot be absorbed, an OLS estimate of β would be biased if $FirmExposure_i$ is correlated with credit demand (Cingano, Manaresi, and Sette 2016; Jiménez et al. forthcoming). Thus, to control for demand for credit, we follow the method developed by Abowd, Kramarz, and Margolis (1999) and recently applied by Bonaccorsi di Patti and Sette (2016) and Cingano, Manaresi, and Sette (2016). Specifically, we include in the cross-sectional specification 2 the vector of firm-level fixed effects $\hat{\alpha}_i$ estimated from the within-firm model 1.¹⁹ As in Khwaja and Mian (2008), standard errors are clustered at the main bank level—that is, the institution that a certain firm has the highest percentage of borrowing with before the shock.

3. Data and Descriptive Statistics

We merge four unique data sets held and managed by the Portuguese Central Bank: (i) the Central Credit Register (Central de Responsabilidades de Crédito), (ii) Individual Information on Interest Rates (Informação Individual de Taxas de Juro), (iii) Central Balance Sheet Database (Central de Balanços), and (iv) the Bank Supervisory Database.

The credit register provides confidential information on all credit exposures above €50, alleviating any concerns regarding unobserved changes in bank credit to SMEs. It covers loans granted to nonfinancial companies by all banks operating in Portugal since reporting is mandatory. Besides recording the total outstanding debt of every firm with each bank at the end of every quarter, each claim specifies the amount that each borrower owes the bank in the short and long-term and the amount that is past due. It also provides information

¹⁹ Jiménez et al. (forthcoming) propose an alternative method to correct for the bias that arises if firm exposure to the shock is correlated with credit demand in the firm-level regressions based on a numerical correction exploiting the difference between OLS and FE estimates of β in the Khwaja and Mian (2008) within-firm regression. Cingano, Manaresi, and Sette (2016) show that the approach of Jiménez et al. (forthcoming) and that we use in this paper are equivalent.

on other credit characteristics such as the undrawn amount of a credit line or credit card.

The database on Individual Information on Interest Rates reports matched firm–bank interest rate information on new loans. While only banks with an annual volume of new corporate loans of more than €50 million were required to report between June 2012 and December 2014, this requirement was extended to all resident banks in January 2015. For consistency, we restrict the analysis to banks that reported interest rate information before and after this reporting change. Besides interest rates, we have loan-level information on the amount, maturity, date of origination, whether the loan is collateralized, and loan type.

The Central Balance Sheet Database provides detailed financial information with an annual frequency for virtually all Portuguese firms—including total assets, incorporation year, equity, net income, number of employees, total debt, and cash holdings. Finally, we also match the above data sets with bank balance sheet data from the Bank Supervisory Database. Given the very low threshold to capture credit exposures in the credit register and the compulsory reporting of balance sheet information by all firms and banks operating in Portugal, the combined firm–bank loan-level data set we use is arguably one of the most comprehensive available in advanced economies.

Table 1 presents firm-level descriptive statistics computed using the matched firm–bank sample. Specifically, we present the mean, median, and standard deviation of the dependent variables as well as firm and bank characteristics across the 40,927 firms and 98 banks in our sample. On average, firms' total committed credit and granted credit lines increased 1.1% and 0.3%, respectively, from the pre- (2013:Q4–2014:Q2) to the postshock period (2014:Q3–2015:Q3). Over the same period, firm investment shrank between 1.6% and 2.6%, employment increased between 2.5% and 3.2%, and cash holdings increased 10.8%. Finally, there was an average decrease in interest rates from the pre to the postresolution period of 94 and 83 basis points on total credit and credit lines, respectively, an increase in loan maturity of 1.8 months, and a decrease in the share of collateralized credit of 4.2 percentage points.

Turning to firm characteristics, the average prefailure firm exposure to the bail-in was 0.008, with a standard deviation of 0.013—the average bank exposure to the bail-in across the 116,245 firm–bank relationships is 0.009, with a standard deviation of 0.020. Firms in our sample have on average four lending relationships, and 32% started a new lending relationship within a year after the resolution. SMEs constitute 98% of all firms. Before the shock, the average firm had €0.75 million in assets, was operating for 13.6 years, had a capital ratio of 26%, suffered losses of 0.6% of total assets, and had a current ratio of 2.2. Finally, we also present bank characteristics, which are averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank.

Table 1
Summary statistics

Variables	N	Mean	Median	SD
<i>Dependent variables</i>				
$\Delta \log$ total credit	40,927	0.011	−0.031	0.485
$\Delta \log$ credit lines	14,320	0.003	0.008	0.570
$\Delta \log$ tangible assets	40,927	−0.026	−0.054	0.978
$\Delta \log$ fixed assets	40,927	−0.016	−0.049	0.959
$\Delta \log$ employees	40,927	0.032	0.000	0.433
$\Delta \log$ worked hours	40,927	0.025	0.009	0.646
$\Delta \log$ cash holdings	40,927	0.108	0.117	1.526
Δ interest rate on total credit	17,632	−0.942	−1.006	3.108
Δ interest rate on credit lines	5,834	−0.831	−0.815	2.513
Δ maturity	17,632	1.830	0.000	22.95
Δ collateral	17,632	−0.042	−0.014	0.280
<i>Firm characteristics</i>				
Firm exposure	40,927	0.008	0.002	0.013
No. bank relationships	40,927	4.106	3.000	2.280
New lending relationship	40,927	0.323	0.000	0.467
SME	40,927	0.983	1.000	0.129
Firm size	40,927	13.53	13.40	1.516
Firm age	40,927	2.679	2.773	0.752
Firm ROA	40,927	−0.006	0.008	0.143
Firm capital ratio	40,927	0.261	0.286	0.424
Firm current ratio	40,927	2.191	1.414	3.555
<i>Bank characteristics</i>				
Bank size	40,927	23.90	24.36	1.349
Bank ROA	40,927	−0.010	−0.009	0.008
Bank capital ratio	40,927	0.054	0.053	0.021
Bank liquidity ratio	40,927	0.012	0.011	0.005
Bank NPLs	40,927	0.064	0.065	0.020

This table presents firm-level summary statistics computed using the matched firm–bank sample. The firm-specific change in the log level of total committed credit and the change in the log level of granted credit lines are constructed by collapsing (time-averaging) the quarterly data for each credit exposure into a single pre- (2013:Q4–2014:Q2) and postshock (2014:Q3–2015:Q3) period. Log change in investment (tangible or fixed assets), employment (number of employees or total number of worked hours), and cash holdings are the firm-specific changes in the log level of the each variable between 2013:Q4 and 2015:Q4. Change in interest rates on total credit and credit lines (in percentage points), maturity (in months), and share of collateralized credit (in percentage points) refer to the firm-level change in the loan-amount-weighted value of the respective variable. Since the interest rate data set only captures new credit operations rather than outstanding amounts, we consider all new credit operations for each firm between 2013:M12 and 2014:M7 (pre period) and 2014:M9 and 2015:M9 (post period). Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. New lending relationship is a dummy variable taking the value of one if the firm has a new loan after the shock (2014:Q3–2015:Q3) with a bank that it had no loan with before, and zero otherwise. Firm size categories are defined according to the E.U. Recommendation 2003/361. Firm characteristics include size (log of total assets), age ($\ln(1+\text{age})$), ROA (net income to total assets), capital ratio (equity to total assets), and current ratio (current assets to current liabilities)—all measured as of 2013:Q4. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are also measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans).

4. Results

In this section, we first examine the effect of the bank resolution through a bail-in on credit supply and firms' borrowing conditions—at both the intensive

and extensive margins. We then trace these effects to real sector outcomes and examine the role of firms' internal liquidity position in explaining our findings.

4.1 Within-firm analysis

4.1.1 Bank resolution and credit supply. The within-firm results in Table 2 show that firms more exposed to the bail-in suffered a significant supply-driven contraction of credit at the intensive margin. Columns 1 and 2 present the results without and with bank-level controls measured as of 2013:Q4—bank size, ROA, capital ratio, liquidity ratio, and NPLs. Column 3 differentiates the main effect of interest across SMEs and large firms. The unit of observation is the change in the log level of total committed credit between each of the 116,245 firm–bank pairs, corresponding to 40,927 firms and 98 banks. All specifications focus on borrowers with multiple bank relationships and include firm fixed effects, ensuring that any observed changes in lending are supply driven and not the result of idiosyncratic firm-level shocks such as changes in credit demand or in the risk profile of a borrower.

The relative credit contraction is both statistically and economically significant. Specifically, the coefficient of interest in column 2 indicates that a one–standard deviation increase in bank exposure to the bail-in (0.020) is associated with a supply-driven decrease in total credit for the average firm of 3.0%. While the effect is significant across the different firm size groups, the results in column 3 show that it is more than twice as strong for large firms than for small- and medium-sized firms—6.3% versus 2.9%, respectively.

In columns 4 to 6 of Table 2, we focus on firms with credit lines granted by multiple banks. This corresponds to 14,320 out of 40,927 firms, for a total of 39,573 firm–bank relationships. In line with Ippolito et al. (2016), who find that following the 2007 freeze of the European interbank market Italian banks managed liquidity risks by extending fewer and smaller credit lines, the coefficient estimates show that granted credit lines were the main channel through which the bail-in led to a reduction in credit supply—a decrease of 5.7% to the average firm for a one–standard deviation increase in bank exposure to the shock.²⁰

4.1.2 Identifying assumptions. The validity of our identification strategy relies on two main assumptions. First, our quasi-experimental research design requires that in the absence of treatment (i.e., the bank failure and subsequent resolution), all banks would have a similar trend in terms of supply of credit. While the parallel trends assumption cannot be tested explicitly due to the absence of a counterfactual, Figure 2 shows it is likely to be satisfied in our

²⁰ The differential effects between total committed credit and granted credit lines are not driven by sample differences. Comparing the results in columns 2 and 4 corresponding to the full and restricted samples using the change in the log level of total committed credit as outcome variable, we observe that the coefficient estimates are not only both negative and statistically significant, but also have the same magnitude.

Table 2
Bank exposure to the bail-in and credit supply – baseline results

	$\Delta \log$ $TotalCredit_{bi}$			$\Delta \log$ $TotalCredit_{bi}$	$\Delta \log$ $CreditLines_{bi}$	
	(1)	(2)	(3)	(4)	(5)	(6)
Bank exposure	−0.023*** (−3.570)	−0.030* (−1.844)		−0.030*** (−4.751)	−0.057*** (−3.156)	
Bank exposure × SMEs			−0.029* (−1.739)			−0.055*** (−3.017)
Bank exposure × large firms			−0.063*** (−3.746)			−0.084*** (−4.676)
No. observations	116,245	116,245	116,245	39,573	39,573	39,573
No. firms	40,927	40,927	40,927	14,320	14,320	14,320
No. banks	98	98	98	95	95	95
Adj. R^2	.047	.049	.050	.065	.103	.103
Bank controls	N	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y
Credit lines with > 1 banks	N	N	N	Y	Y	Y

This table presents estimation results of the within-firm specification 1, where the dependent variables are the change in the log level of total committed credit or granted credit lines between each firm–bank pair. The quarterly data for each credit exposure are collapsed (time-averaged) into a single pre- (2013:Q4–2014:Q2) and postshock period (2014:Q3–2015:Q3). Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm size categories are defined according to the E.U. Recommendation 2003/361. All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

setting. In detail, we use a modified version of the within-firm specification 1, regressing for each quarter the change in the log level of total committed credit between each firm–bank pair in that quarter relative to 2014:Q2 (the last period before the shock) on bank exposure and firm fixed effects. The dashed lines indicate the 5%–95% confidence interval using standard errors clustered at the bank level. Before the shock, there is no significant variation in credit provision across banks. However, credit supply decreased significantly starting in 2014:Q3 and deteriorated over time as a result of the bail-in.

Second, the implicit assumption behind using firm fixed effects to control for idiosyncratic demand shocks in the Khwaja and Mian (2008) within-firm specification is that loan demand changes proportionally across all banks lending to the firm—that is, individual firms take their multiple banks as providers of a perfectly substitutable good. In our setting, this assumption could be violated if, for instance, firms reduced credit demand from the bailed-in bank after the shock while increasing it from other banks. However, some factors suggest the effects we observe are indeed supply driven and unlikely to be explained by within-firm changes in demand for overall credit or credit lines in particular. First, changes to the management and governance structure of the bailed-in bank effectively made it more risk-averse in its lending policies.

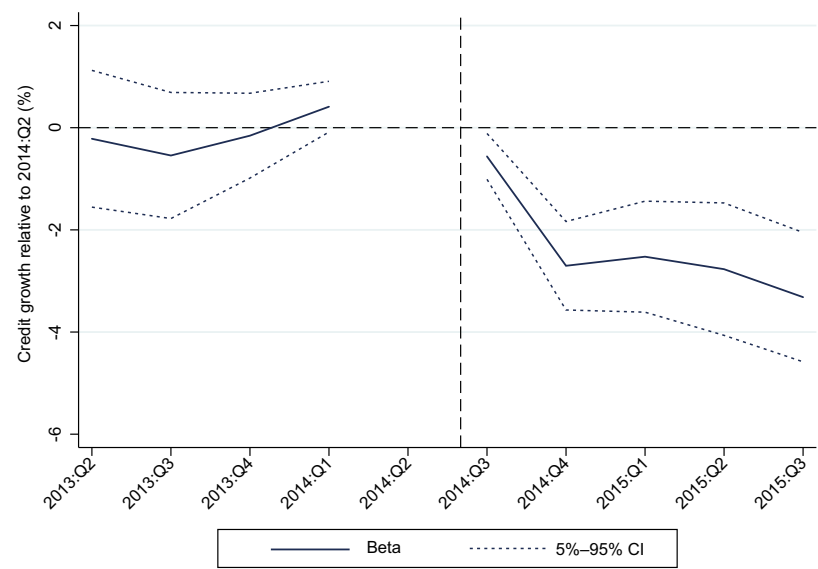


Figure 2
Impact of the bail-in on credit supply at the intensive margin

This figure presents coefficient estimates of a modified version of the within-firm specification 1 where the dependent variable ($\Delta \log(Credit)_{bi}$) is regressed on $BankExposure_b$ and firm fixed effects. Each coefficient estimate in each quarter corresponds to a different within-firm regression, where the outcome variable is the change in the log level of total committed credit between each firm–bank pair between the respective quarter and 2014:Q2. All coefficients are scaled by the corresponding variable’s standard deviation. The dashed lines indicate the 5%–95% confidence interval, with standard errors clustered at the bank level.

In fact, as stated in both its 2014 and 2015 annual reports, the contraction in corporate loans was achieved mainly through the reduction and nonrenewal of credit lines as well as a decrease in its credit exposure to large firms, with any new large loan requiring extensive review and risk committee board approval (Novo Banco 2014). The bank’s strategy to manage its capital and liquidity positions prudently by deleveraging noncore assets from the loan portfolio and optimizing risk-weighted assets led to a reduction of its balance sheet of 18% from the day of the resolution until the end of 2015 (Novo Banco 2015). Given that the bailed-in bank is by far the most exposed to the resolution (i.e., it has the highest bank exposure value as shown in Table IA1 in the Internet Appendix), the restructuring plan following the intervention focused on reducing large exposures and credit lines is entirely consistent with the credit supply contraction at the intensive margin we show.

In addition, in contrast with a shift in firm demand from the bailed-in bank to other banks explained by reputational damage or even liquidity and solvency concerns, the 13% contraction in corporate loans from August 2014 to December 2015 was accompanied by a 7.4% increase in customer deposits (Novo Banco 2015). This suggests that despite the challenges brought by the

resolution measure, the bank was able to stabilize its funding sources and, at least partially, recover its customers' confidence.

Finally, as highlighted by Paravisini, Rappoport, and Schnabl (2020), our identifying assumption may also be violated if the bailed-in bank was specialized in certain industries or sectors such as export markets. In such segments where some banks may have more expertise than others, credit is no longer a homogeneous good offered across different banks and, as a result, sector-level demand shocks may ultimately lead to firm–bank-specific loan demand. Nevertheless, untabulated results (for confidentiality reasons) suggest that firm–bank-specific demand due to sector specialization is not a source of concern in our setting. In fact, the bailed-in bank was active in all main industries and did not control the majority of the lending activity in any of them. Our results could also be biased if certain banks were targeting their lending to firms in industries experiencing particularly severe (and correlated) demand-side shocks. However, when we compare the relative importance of certain industries for the bailed-in bank vis-à-vis all other banks, we observe no discernible differences across industries between the two groups.

4.1.3 Robustness tests. The within-firm results presented above are robust to several important tests. First, we use an alternative bank exposure measure based on daily five-year CDS spreads on senior unsecured debt, considering the four banks operating in Portugal that are classified as significant institutions by the ECB for which there are available CDS spread data. In detail, we define bank exposure to the shock as the bank-specific increase in CDS spreads from 2014:Q2 (one month before the resolution) to 2014:Q3 (two months after the resolution), as illustrated by the light gray area in Figure 1. This indicator measures the market's perception of the default probability increase for the resolved bank as well as the magnitude of potential spillovers for the three other major Portuguese banks following the bail-in. Consistent with the estimates in the baseline regressions, the results in columns 1 to 3 of Table IA2 in the Internet Appendix show that a one-standard deviation increase in bank exposure to the shock captured through the reaction of CDS spreads (0.007) leads to a decrease in the supply of credit of 3.00% for the average firm—2.60% for SMEs and 8.20% for large firms.²¹

Second, to ensure our results are not confined to firms with multiple bank relationships, we use the complete sample of borrowing firms in Portugal, including single-bank-relationship firms, in a model that replaces firm fixed effects with industry-location-size fixed effects following Degryse et al. (2019). The group contains only the firm itself in case the firm has multiple lending

²¹ Table IA1 in the Internet Appendix reports the values of this alternative bank exposure measure for each bank. Since there are only four banks with available CDS spread data, standard errors clustered at the bank level would be biased. Thus, in columns 1 to 3 of Table IA2, we use heteroskedasticity-consistent standard errors instead. The coefficients of interest are still significant at conventional levels when using the wild cluster bootstrap method of Cameron, Gelbach, and Miller (2008).

relationships, while firms with single-bank relationships are grouped based on the district in which they are headquartered, their industry, and deciles of loan size. The results are reported in columns 4 to 6 of Table IA2. Despite the considerable increase in the number of firms (from 40,927 to 85,216), the coefficient estimates are remarkably similar to those in the smaller sample restricted to firms with multiple bank relationships, both in terms of magnitude and statistical significance.

Third, we limit our sample to loan operations and thus disregard both used and unused credit lines (Table IA3, columns 1–3). In this case, only the coefficient estimate for large firms enters significantly and is smaller in magnitude when compared to the estimate for total credit. This confirms that the shock led to a reduction in credit supply primarily through fewer and smaller credit lines.

Finally, in columns 4 to 6 of Table IA3 we follow Iyer et al. (2014) and compare lending immediately before (2014:Q2) and one year after the shock (2015:Q3) instead of time-averaging the quarterly credit exposures into a pre- (2013:Q4–2014:Q2) and postresolution (2014:Q3–2015:Q3) period. Our results are the same, if not stronger, when compared to our baseline regressions.

4.1.4 Firm heterogeneity. While we observe a reduction in credit supply on average and particularly for larger firms, this contraction might vary across other firm characteristics. In this respect, the results in Table 3 examine whether there is further cross-sectional variation in the effect of the resolution by introducing an interaction term between bank exposure and several firm-level characteristics measured before the shock. Columns 1 to 7 show there are no significant differential effects along any cross-sectional dimension considered, including firm age, capital, profitability, liquidity, or riskiness—measured by the interest coverage ratio, the share of NPLs in total credit, or when splitting firms according to whether they had negative equity before the resolution, a common indicator to capture “zombie firms” (e.g., Bonfim et al. 2020).

Instead, the results in column 8 show a negative and significant interaction term of bank exposure with a dummy equal to one if the firm’s main lender before the shock was the bailed-in bank. While this result contrasts with the evidence on the insulating effect of relationship banking on the quantity of credit following negative bank shocks (Bolton et al. 2016; Beck et al. 2018), it highlights the disruptive effect an intervention can have on established firm–bank relationships.²² In contrast, the interaction between bank exposure with a dummy equal to one if the firm’s main lender before the shock was not the bailed-in bank is positive and significant, irrespective of whether the main bank participated in the *ad hoc* loan to the Portuguese Resolution Fund (column 9) or not (column 10). This suggests that the credit supply contraction at the intensive margin after the resolution is solely driven by the bailed-in bank. In other words,

²² In a different setting, Carvalho, Ferreira, and Matos (2015) find that bank distress is associated with equity valuation losses and investment cuts to firms with the strongest lending relationships.

Table 3
Bank exposure to the bail-in and credit supply – firm heterogeneity

	$\Delta \log TotalCredit_{it}$									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bank exposure	-0.031 (-1.517)	-0.033** (-2.029)	-0.030* (-1.845)	-0.031* (-1.847)	-0.032* (-1.935)	-0.031* (-1.875)	-0.031* (-1.848)	-0.009 (-0.529)	-0.059*** (-3.167)	-0.033** (-2.037)
Bank exposure $\times$ firm age	0.000 (0.139)									
Bank exposure $\times$ firm capital ratio		0.010 (1.446)								
Bank exposure $\times$ firm ROA			-0.018 (-0.776)							
Bank exposure $\times$ firm current ratio				0.000 (0.465)						
Bank exposure $\times$ firm interest coverage ratio					0.000 (0.506)					
Bank exposure $\times$ firm share of nonperforming loans						0.043 (1.063)				
Bank exposure $\times$ firm has negative equity							0.003 (0.387)			
Bank exposure $\times$ firm main lender is the bailed-in bank								-0.063*** (-7.853)		
Bank exposure $\times$ firm main lender participated in <i>ad hoc</i> loan									0.048*** (6.005)	
Bank exposure $\times$ firm main lender did not participate in <i>ad hoc</i> loan										0.034** (2.270)
No. observations	116,245	116,245	116,245	116,245	116,245	116,245	116,245	116,245	116,245	116,245
No. firms	40,927	40,927	40,927	40,927	40,927	40,927	40,927	40,927	40,927	40,927
No. banks	98	98	98	98	98	98	98	98	98	98
Adj. R^2	.049	.049	.049	.049	.049	.050	.049	.051	.050	.050
Bank controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

This table presents estimation results of the within-firm specification 1 with bank exposure interacted with several firm-level characteristics measured before the shock. The dependent variable is the change in the log level of total committed credit between each firm–bank pair. The quarterly data for each credit exposure are collapsed (time-averaged) into a single pre- (2013:Q4–2014:Q2) and postshock period (2014:Q3–2015:Q3). Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm assets, number of employees, age, capital ratio, ROA, current ratio, and interest coverage (gross profit over interest expense on loans) are all measured as of 2013:Q4. All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

we find no evidence of a credit supply contraction for firms whose main lender was any other bank operating in Portugal.

4.1.5 Effects on credit conditions. We have so far focused on the consequences of the bank bail-in on the quantity of credit. Nevertheless, the resolution may have also impacted the interest rates charged on loans and credit lines, the maturity on the credit granted, or the amount of collateral required. Santos (2011) and Chodorow-Reich (2014), for instance, show using syndicated loan data that firms that had relationships with less healthy lenders during the financial crisis paid relatively higher interest rates afterwards. To examine the effects of the resolution on credit conditions, we estimate the same within-firm specification 1 using the change in interest rates, maturity, and share of collateralized credit between 2013:M12 and 2014:M7 (pre period) and 2014:M9 and 2015:M9 (post period) as outcome variables.

The results reported in Table 4 show a significant increase in interest rates charged on new loans and credit lines, as well as a considerable tightening of loan maturities and required collateral after the bail-in. Compared to Table 2, we now also control for other preperiod loan characteristics available in this restricted sample such as the preshock, loan-amount-weighted maturity and share of collateralized credit for a bank–firm relationship when examining the effect on interest rates. In detail, the coefficient estimates in columns 1 and 3 indicate that a one–standard deviation increase in bank exposure to the bail-in is associated with an increase in interest rates on all new credit operations and granted credit lines for the average firm of 0.40 and 1.54 percentage points, respectively. This corresponds to a 4.56% (18.07%) increase relative to the mean interest rate on all new loans (granted credit lines) before the shock. Similarly, the results in columns 5 and 7 show that a one–standard deviation increase in bank exposure to the resolution is associated with a decrease in loan maturity of 7.51% relative to the preshock mean and an increase in the share of collateralized credit of 25.8 percentage points. Apart from loan maturity, where the effect is concentrated on SMEs, the tightening of credit conditions following the resolution is statistically significant and economically large irrespective of the firm size.²³

²³ While most credit registers in advanced economies do not contain this type of granular information on interest rates, some limitations should be noted—in line with those highlighted by Cingano, Manaresi, and Sette (2016) for the case of Italy, for instance. First, the data set contains interest rate, maturity, and collateral information on new credit operations rather than on total outstanding credit. Second, since the requirement that only banks with an annual volume of new corporate loans of more than €50 million have to report was only relaxed in 2015, these data are only available for the largest 11 banks. Thus, given that the sample is restricted to firms drawing new credit from at least two of these 11 banks, there is a noticeable drop in the sample size relative to the within-firm credit quantity regressions. Importantly, however, the database on Individual Information on Interest Rates has the same firm and bank identifiers as the credit register and includes both the bailed-in bank and all the banks that participated in the *ad hoc* loan to the Portuguese Resolution Fund. Finally, standard errors clustered at the bank (or main bank) level would likely be biased given that the relatively few reporting banks. As a result, we use heteroskedasticity-consistent standard errors instead. As in columns 1 to 3 of Table IA2 in the Internet Appendix, our conclusions do not change when using the wild cluster bootstrap method of Cameron, Gelbach, and Miller (2008).

Table 4
Bank exposure to the bail-in and credit conditions

	$\Delta Interest Rates$ <i>AllNewCreditOperations</i> _{bi}		$\Delta Interest Rates$ <i>CreditLinesOnly</i> _{bi}		$\Delta Maturity$ _{bi}		$\Delta Collateral$ _{bi}	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bank exposure	0.397*** (7.929)		1.537*** (8.683)		-0.521** (-2.120)		0.258*** (69.622)	
Bank exposure × SMEs		0.395*** (7.871)		1.575*** (8.852)		-0.500*** (-2.025)		0.258*** (69.271)
Bank exposure × large firms		0.434*** (2.902)		0.948* (1.815)		-0.937 (-1.325)		0.272*** (29.865)
No. observations	49,623	49,623	13,991	13,991	49,623	49,623	49,623	49,623
No. firms	17,632	17,632	5,834	5,834	17,632	17,632	17,632	17,632
No. banks	11	11	9	9	11	11	11	11
Adj. R ²	.164	.164	.125	.127	.117	.117	.255	.255
Preshock loan interest rate	N	N	N	N	Y	Y	Y	Y
Preshock loan maturity	Y	Y	Y	Y	N	N	Y	Y
Preshock loan collateral	Y	Y	Y	Y	Y	Y	N	N
Bank controls	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y	Y	Y

This table presents estimation results of the within-firm specification 1, where the dependent variables are the change in the loan-amount-weighted interest rates on new credit operations, interest rates on credit lines, maturity, and share of collateralized credit between each firm–bank pair. Since the interest rate data set only captures new operations (rather than outstanding amounts), we consider all new credit operations between a firm and a bank between 2013:M12 and 2014:M7 (pre period) and 2014:M9 and 2015:M9 (post period) when computing these measures. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm size categories are defined according to the E.U. Recommendation 2003/361. All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

4.2 Cross-sectional analysis

4.2.1 Bank resolution and credit supply. So far we have gauged the effect of bank resolution at the intensive margin. However, the within-firm estimations ignore credit flows from new lending relationships as well as bank relationships that were terminated from the pre- to the postresolution period. Therefore, we now turn to the cross-sectional (between-firm) estimations. Since we cannot use firm-fixed effects in the regressions analyzing the overall impact of bank shocks on credit supply, we control for omitted firm-level factors such as credit demand with a two-step estimation based on Abowd, Kramarz, and Margolis (1999). Specifically, we include in the estimations the vector of firm-level dummies estimated in column 1 of Table 2.²⁴ We also include industry and district fixed effects as additional controls for unobservable demand and risk profile differences.

The results in Table 5 show that, on average, there was no decrease in overall credit after the shock for firms more exposed to the bail-in when compared to firms exposed less, including when differentiating between firms of different sizes. However, we do observe a binding contraction in credit lines for SMEs more exposed to the resolution. In detail, the explanatory variable of interest, firm exposure, is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Columns 1 and 2 report the results on total committed credit for the average firm without and with firm and bank controls, respectively, while column 3 differentiates between SMEs and large enterprises. None of the coefficients enters significantly at conventional levels when including the full set of controls. In columns 4 to 6 we instead focus on firms with credit lines granted by multiple banks. As in the within-firm estimations reported in Table 2, this corresponds to 14,320 out of 40,927 firms. Comparing the results in columns 2 and 4 corresponding to the full and restricted samples using the change in the log level of total committed credit as the outcome variable, we observe that the coefficient estimates are statistically insignificant and have a similar magnitude, confirming again that the differential effects between total committed credit and granted credit lines are not driven by sample differences. More importantly, the results in columns 5 and 6 indicate that SMEs more exposed to the shock suffered a considerable decrease in the amount of credit lines available to them. Specifically, for a one-standard deviation in firm exposure to the bail-in, SMEs experienced a 2.2% binding decrease in granted credit lines.

4.2.2 Robustness tests. The results above are robust to several tests. First, we consider an alternative firm exposure measure that, as in Table IA2, is based on

²⁴ If biases due to endogenous matching between firms and banks were present in our data, we should observe a substantial correlation between exposure and $\hat{\alpha}_i$ (Cingano, Manaresi, and Sette 2016; Jiménez et al. forthcoming). However, exploiting model 1, we find that the estimated vector of firm-level dummies is virtually uncorrelated with our main bank exposure measure ($\rho = 0.0014$).

Table 5
Firm exposure to the bail-in and credit supply

	$\Delta \log$ <i>TotalCredit_i</i>			$\Delta \log$ <i>TotalCredit_i</i>	$\Delta \log$ <i>CreditLines_i</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm exposure	-0.010* (-1.983)	-0.005 (-0.901)		-0.004 (-0.987)	-0.022*** (-4.103)	
Firm exposure × SMEs			-0.005 (-0.936)			-0.022*** (-4.316)
Firm exposure × large firms			-0.003 (-0.287)			-0.007 (-0.463)
No. observations/firms	40,927	40,927	40,927	14,320	14,320	14,320
No. banks	98	98	98	95	95	95
Adj. R^2	.363	.378	.378	.393	.175	.175
Firm controls	N	Y	Y	Y	Y	Y
Bank controls	N	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y
No. bank relationships >1	Y	Y	Y	Y	Y	Y
Credit lines with >1 banks	N	N	N	Y	Y	Y

This table presents estimation results of cross-sectional model 2, where the dependent variables are the change in the log level of total committed credit or granted credit lines for each firm. The quarterly data for each credit exposure are collapsed (time-averaged) into a single pre- (2013:Q4–2014:Q2) and postshock period (2014:Q3–2015:Q3). Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm-level controls, defined in Table 1, are also measured in 2013:Q4. Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

the reaction of CDS spreads on senior unsecured debt and considers the four banks operating in Portugal that are classified as significant institutions by the ECB. In detail, in columns 1 to 4 of Table IA4 in the Internet Appendix, firm exposure is computed as the weighted average of bank exposure across all banks lending to a firm (using as weights the preperiod share of total credit from each bank), but where bank exposure to the shock is defined as the bank-specific increase in CDS spreads from 2014:Q2 (one month before the resolution) to 2014:Q3 (two months after the resolution)—thus capturing the market's perception of the default probability increase for the resolved bank as well as the magnitude of potential spillovers for the three other major Portuguese banks following the bail-in. Second, in columns 5 to 8 of Table IA4, we follow the same procedure but define bank exposure as a dummy variable equal to one for the bailed-in bank and zero otherwise—that is, we implicitly assume that only the bailed-in bank was affected by the resolution and there were no spillover effects to other banks in the system. In either case, our conclusions remain

the same, with the latter test suggesting that the effects we observe are indeed driven exclusively by the bailed-in bank.

Finally, we also confirm our findings when comparing lending immediately before (2014:Q2) and one year after the shock (2015:Q3) instead of time-averaging the quarterly credit exposures into a pre- and postresolution period (Table IA5, columns 1–2), and when extending the sample to all firms, including those with only one lending relationship (Table IA5, columns 3–4). In the latter specifications, credit demand is the vector of firm-level dummies estimated from the within-firm regression with industry-location-size fixed effects as in Table IA2. Together, these tests confirm that while more exposed firms were able to compensate for the tightening of overall credit with other sources of funding, SMEs were subject to a binding contraction of funds available through credit lines.

4.2.3 Role of lending relationships and effects on credit conditions. The disruption of established bank–firm relationships can ultimately have negative effects on real outcomes, particularly if borrowers are unable to replace these relationships with other lenders on equal terms (Bernanke 1983; Ashcraft 2005).

We investigate this issue in Table 6 where the dependent variables are now either a dummy that takes a value of one if a firm takes out a loan from a bank with which it had no lending relationship before the resolution or a dummy that takes a value of one if a firm kept the relationship with a bank after the resolution (i.e., the firm kept at least one outstanding loan with the bank) but the bank terminated a credit line that was available to the firm before the shock. The estimates in columns 1 and 2 show that firms more exposed to the bail-in were as likely to start a new lending relationship after the shock as firms exposed less. This result can be explained by the fact that the average firm already had four bank relationships before the shock (Table 1). In addition, the results in columns 3 and 4 indicate that, in line with what the bailed-in bank stated following the resolution (Novo Banco 2014, 2015), large firms more exposed to the bail-in were more likely to have its credit line not renewed after the shock when compared to large firms exposed less. This finding therefore suggests the bailed-in bank did break with the former legacy by implementing a more sustainable lending policy.²⁵

Notably, the coefficient estimates reported in columns 5 to 6 of Table 6 confirm that the eight (large) banks that had to contribute to the *ad hoc* loan to the Portuguese Resolution Fund (i.e., those banks that were less but still exposed to the resolution) were crucial for firms to maintain credit. Specifically, the dependent variable is now the change in the log level of total committed credit

²⁵ The results in columns 5 and 6 of Table IA5 show that more exposed SMEs were relatively less likely to terminate existing lending relationships, highlighting the challenges for smaller firms more reliant on relationship lending to replace existing bank relationships due to the time it takes to build up soft information.

Table 6
Lending relationships and credit supply

	<i>New Lending Relationship_i</i>		<i>Nonrenewal of Credit Line_i</i>		<i>ΔlogCredit_i (Except Bailed-in Bank)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm exposure	−0.008 (−1.442)		−0.001 (−0.257)		0.060*** (9.621)	
Firm exposure × SMEs		−0.009 (−1.464)		−0.002 (−0.458)		0.059*** (9.269)
Firm exposure × large firms		−0.003 (−0.527)		0.020** (2.217)		0.070*** (8.097)
No. observations/firms	40,927	40,927	40,927	40,927	40,927	40,927
No. banks	98	98	98	98	98	98
Adj. <i>R</i> ²	.058	.058	.023	.023	.342	.342
Firm controls	Y	Y	Y	Y	Y	Y
Bank controls	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y

This table presents estimation results of cross-sectional model 2, where the dependent variables are a dummy that takes the value of one if a firm takes out a loan from a bank with which it had no lending relationship before the shock, a dummy that takes the value of one if a firm terminates an existing lending relationship with a bank after the resolution, and the change in the log level of total (committed) credit for each firm between 2013:Q4 and 2015:Q3 excluding the bailed-in bank. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age ($\ln(1+\text{age})$), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

to each firm from all banks except the bailed-in bank from the pre- (2013:Q4–2014:Q2) to the postresolution period (2014:Q3–2015:Q3). The results show a significant and positive relationship between firm exposure and credit growth, with a one-standard deviation increase in firm exposure to the bail-in being associated with a 6.0% increase in lending from these other banks.

Finally, turning to the cross-sectional effect of the resolution on credit conditions, the results in columns 1 to 4 of Table 7 show that firms across all size groups that were more exposed to the bail-in saw a moderate increase in their interest rates on total credit, while only more exposed SMEs suffered a moderate increase in interest rates on granted credit lines. However, the economic effect is modest—a one-standard deviation increase in firm exposure to the bail-in is associated with a 7 bps and 16 bps increase in the interest rates on total credit and credit lines for the average firm, respectively. These results are consistent

Table 7
Firm exposure to the bail-in and credit conditions

	$\Delta \text{Interest Rates}$ $\Delta \text{AllNewCreditOperations}_i$		$\Delta \text{Interest Rates}$ CreditLinesOnly_i		$\Delta \text{Maturity}_i$		$\Delta \text{Collateral}_i$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Firm exposure	0.071** (1.972)		0.164*** (3.218)		-0.595** (-2.148)		0.043*** (13.690)	
Firm exposure $\times$ SMEs		0.066* (1.803)		0.164*** (3.214)		-0.563** (-2.026)		0.044*** (13.679)
Firm exposure $\times$ large firms		0.212*** (3.164)		0.175 (0.986)		-1.438* (-1.813)		0.037*** (4.301)
No. observations/firms	17,632	17,632	5,834	5,834	17,632	17,632	17,632	17,632
No. banks	11	11	9	9	11	11	11	11
Adj. R^2	.110	.110	.066	.066	.035	.035	.078	.078
Preshock loan interest rate	N	N	N	N	Y	Y	Y	Y
Preshock loan maturity	Y	Y	Y	Y	N	N	Y	Y
Preshock loan collateral	Y	Y	Y	Y	Y	Y	N	N
Firm controls	Y	Y	Y	Y	Y	Y	Y	Y
Bank controls	Y	Y	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y	Y	Y

This table presents estimation results of cross-sectional model 2, where the dependent variable is the firm-specific change in the loan-amount-weighted interest rates on new credit operations, interest rates on credit lines, maturity, and share of collateralized credit. Since the interest rate data set only captures new operations (rather than outstanding amounts), we consider all new credit operations between a firm and a bank between 2013:M12 and 2014:M7 (pre period) and 2014:M9 and 2015:M9 (post period) when computing these measures. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Firm size categories are defined according to the E.U. Recommendation 2003/361. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capitalization (regulatory capital ratio), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age (ln(1+age)), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

with Khwaja and Mian (2008) and Cingano, Manaresi, and Sette (2016) who find that bank-level shocks have no meaningful effects on interest rates charged to firms in Pakistan and Italy, respectively. In line with a modest tightening of interest rates, the results in columns 5 and 6 of Table 7 show a statistically significant but economically small reduction in loan maturity across all firms, with a one-standard deviation increase in firm exposure resulting in less than a month's reduction in loan maturity. We also find a relatively small increase in the share of collateralized credit after the shock (columns 7 and 8).

Overall, the cross-sectional analysis so far shows that both SMEs and large firms that were more exposed to the bail-in did not suffer an overall reduction in credit when compared to firms exposed less. In fact, these firms were able to compensate for the reduction in credit at the intensive margin with lending from other financial institutions they already had a relationship with. Furthermore, despite being statistically significant, the effects on credit conditions are economically small and thus unlikely to be driving by themselves any potential changes in real outcomes. However, when isolating credit lines from total committed credit by focusing on firms with multiple credit lines, we show that SMEs more exposed to the resolution were subject to a binding contraction in quantity of funds available through lines of credit—a crucial component for corporate liquidity management and the dominant source of liquidity for firms around the world (e.g., Sufi 2009; Jiménez, Lopez, and Saurina 2009).

4.3 Real effects

4.3.1 Impact on investment and employment. What was the effect of changes in financing conditions on investment and employment decisions taken by the affected firms? On the one hand, it is not clear that we should find significant real effects given the continued access to the same overall level of external funding, though with smaller granted credit lines for SMEs. On the other hand, the results point towards higher uncertainty for more exposed firms, as they had to compensate for the lost funding at the intensive margin with credit from other banks and (re)negotiate loan terms and conditions. We therefore turn to investment and employment growth as real sector outcome variables, and then focus on the role of firms' internal liquidity in driving the results.

The results in Table 8 show a relative reduction in investment for SMEs that were more exposed to the resolution. The dependent variable in columns 1 to 3 is the change in the log level of tangible assets for each firm between 2013:Q4 and 2015:Q4, with column 1 presenting a regression for the 14,320 firms with multiple credit lines at different banks and columns 2 and 3 focusing on our main sample of 40,927 firms with more than one bank relationship. As before, all specifications include firm and bank controls, a proxy for credit demand, and industry and district fixed effects. In both cases, firm exposure enters negatively and significantly. This reduction in investment, however, is

Table 8
Firm exposure to the bail-in and investment

	<i>ΔlogTangibleAssets_i</i>			<i>ΔlogFixedAssets_i</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
Firm exposure	−0.020*** (−3.611)	−0.019*** (−4.838)		−0.015*** (−2.565)	−0.013*** (−3.112)	
Firm exposure × SMEs			−0.020*** (−4.766)			−0.013*** (−2.983)
Firm exposure × large firms			−0.006 (−0.718)			−0.006 (−0.577)
No. observations/firms	14,320	40,927	40,927	14,320	40,927	40,927
No. banks	95	98	98	95	98	98
Adj. <i>R</i> ²	.045	.041	.041	.043	.039	.039
Firm controls	Y	Y	Y	Y	Y	Y
Bank controls	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y
Credit lines with > 1 banks	Y	N	N	Y	N	N

This table presents estimation results of cross-sectional model 2, where the dependent variables are the change in the log level of tangible assets and in the log level of fixed assets for each firm between 2013:Q4 and 2015:Q4. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age ($\ln(1+\text{age})$), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

only significant for SMEs (column 3), with a one-standard deviation increase in firm exposure to the bail-in associated with a 2.0% relative reduction in investment among SMEs. Our results remain the same when using the change in the log level of fixed assets as the dependent variable (Table 8, columns 4–6), our two alternative firm exposure measures as in Table IA4 (Table IA6, columns 1–4), when normalizing the change in tangible assets or fixed assets by the firms' preperiod total assets (Table IA6, columns 5–8), or when including firms with only one bank relationship in the analysis (Table IA7, columns 1–2).

In line with the evidence for investment, columns 1 to 3 of Table 9 show a significant and negative relationship between firm exposure to the bail-in and the growth of the number of employees at firms. As before, this effect is concentrated in SMEs and is not significant for large enterprises.²⁶ Controlling

²⁶ In Table IA8 in the Internet Appendix, we split firms into four size groups according to (i) the E.U. classification (micro, small, medium, and large firms); (ii) total assets (€0–€14.33 million, €14.34–€28.66 million, €28.67–€43 million, >€43 million); and (iii) number of employees (0–83, 84–166, 167–250, >250). Overall, the binding

Table 9
Firm exposure to the bail-in and employment

	$\Delta \log \text{No. Employees}_i$			$\Delta \log \text{No. Worked Hours}_i$		
	(1)	(2)	(3)	(4)	(5)	(6)
Firm exposure	-0.014*** (-2.994)	-0.012*** (-4.116)		-0.019*** (-4.555)	-0.015*** (-3.794)	
Firm exposure $\times$ SMEs			-0.013*** (-4.213)			-0.015*** (-3.820)
Firm exposure $\times$ large firms			-0.002 (-0.488)			-0.004 (-0.713)
No. observations/firms	14,320	40,927	40,927	14,320	40,927	40,927
No. banks	95	98	98	95	98	98
Adj. R^2	.080	.066	.066	.054	.047	.047
Firm controls	Y	Y	Y	Y	Y	Y
Bank controls	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
District FE	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y
Credit lines with > 1 banks	Y	N	N	Y	N	N

This table presents estimation results of cross-sectional model 2, where the dependent variables are the change in the log level of number of employees and in the log level of total number of worked hours for each firm between 2013:Q4 and 2015:Q4. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age ($\ln(1+\text{age})$), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

for firm and bank characteristics, we find a 1.3% relative drop in the number of employees at SMEs for a one-standard deviation increase in exposure to the resolution. To capture different margins of adjustment of employment, we also consider the log change in the total number of hours worked by all firm employees as an alternative outcome variable. The results are reported in columns 4 to 6 of Table 9. The reduction in employment is again only present in more exposed SMEs—a 1.5% relative decrease for a one-standard deviation increase in firm exposure.²⁷

contraction in credit lines and the negative effects on investment and employment are present among micro, small, and medium firms but not large firms irrespective of the definition used.

²⁷ Our findings also remain the same when (i) considering our two alternative firm exposure variables computed as the weighted average of bank exposure across all banks lending to a firm (using as weights the preperiod share of total credit from each bank), but where bank exposure to the shock is defined as the bank-specific increase in CDS spreads from 2014:Q2 to 2014:Q3 (Table IA9, columns 1 and 2) or as a dummy variable equal to one for the bailed-in bank and zero otherwise (Table IA9, columns 3 and 4), and (ii) when including firms with only one bank relationship in the analysis (Table IA7, columns 3–4).

The results in Tables 8 and 9 are particularly important given how crucial SMEs are for the overall Portuguese economy, not only constituting 98% of the firms in our sample but also having 52% of total tangible assets and 57% of total employees—57% and 63%, respectively, if we account for single-bank-relationship firms.²⁸ Nonetheless, our conclusions are in line with Chodorow-Reich (2014) and Berton et al. (2018), who find that smaller firms are particularly vulnerable to the negative impact of a credit crunch on employment. Bottero, Lenzu, and Mezzanotti (2020) also show that while the credit supply contraction in Italy following the European sovereign crisis was similar in magnitude for large and small firms, it led to a reduction in investment and employment only for smaller firms.

4.3.2 Indirect spillover effects. To shed light on the potential aggregate effects of the bail-in, we follow Huber (2018) and examine in Table 10 whether there is evidence of indirect spillover effects for firms operating in districts with a high share of firms exposed to the shock. While these indirect spillover effects would arise through changes in aggregate economic conditions within a district due to the direct effect on affected firms, they would be independent of the individual firms' bank relationships. Specifically, we extend the cross-sectional model 2 as:

$$\Delta \log(Y)_{id} = \beta(\text{FirmExposure}_{id}) + \lambda(\overline{\text{FirmExposure}_{-id}}) + \tau' F_{id} + \delta' \overline{X}_{id} + \hat{\alpha}_{id} + \varepsilon_{id}, \quad (3)$$

where $\Delta \log(Y)_{id}$ is the log change in tangible assets or in the number of employees between 2013:Q4 and 2015:Q4 from all banks to firm i based in district d . $\overline{\text{FirmExposure}_{-id}}$ is the average exposure to the shock across all firms other than firm i operating in the same district d . Unlike Huber (2018), we also include bank controls $\overline{X}_{id}$ averaged at the firm level according to the share of total credit granted to the firm by each bank prior to the shock as well as a control for credit demand ($\hat{\alpha}_{id}$) estimated from the within-firm model 1. In fact, the control variables used in this alternative specification are the same as those in the corresponding Tables 8 and 9, except there are no district fixed effects in columns 1, 2, 4, and 5 of Table 10, and no district and industry fixed effects in columns 3 and 6.

²⁸ To help put into perspective a standard deviation change in our context, consider two identical firms i and k that, before the shock, both had €10 million in total committed credit from BES and CGD, the largest bank in Portugal. Firm i had €5 million in credit from each, and thus an exposure to the shock of 3.49% according to our measure (Table IA1)—50% from BES and 50% from CGD. Given that a one-standard deviation increase in firm exposure corresponds to 1.3% (Table 1), firm k would be one standard deviation apart from firm i if it had €7 million in credit from BES and €3 million from CGD—70% from BES and 30% from CGD. Thus, firm k 's investment and employment after the bail-in would be 2.0 and 1.5% lower than firm i 's, respectively (Tables 8–9).

Table 10
Direct and indirect spillover effects on investment and employment

	$\Delta \log Tangible Assets_i$			$\Delta \log No. Employees_i$		
	(1)	(2)	(3)	(4)	(5)	(6)
Firm exposure	-0.022*** (-5.464)	-0.021*** (-4.862)	-0.022*** (-5.243)	-0.012*** (-3.942)	-0.012*** (-4.190)	-0.013*** (-4.045)
Firm exposure of other firms in the same district		-0.012*** (-2.886)			0.005 (1.599)	
Firm exposure of other firms in the same district and industry			-0.012*** (-3.707)			0.004 (1.066)
No. observations/firms	40,927	40,927	40,927	40,927	40,927	40,927
No. banks	98	98	98	98	98	98
Adj. R^2	.040	.040	.037	.064	.064	.059
Firm and bank controls	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	N	Y	Y	N
District FE	N	N	N	N	N	N
No. bank relationships > 1	Y	Y	Y	Y	Y	Y

This table presents estimation results of cross-sectional model 3. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age ($\ln(1+age)$), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the corresponding within-firm regression. All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

Column 1 shows the direct treatment effect for firm *i*'s investment—a 2.2% decrease in investment for a one-standard deviation increase in firm exposure to the bail-in, in line with the results in Table 8. Column 2, however, shows there are negative spillover effects to other firms in the same district as firm *i*. In detail, if the same firm *i* had operated in a district where the overall exposure to the bail-in of the other firms had been one standard deviation greater, investment of this firm *i* would have fallen by a total of 3.3%. In this district, firms not exposed to the bail-in at all would still reduce investment by 1.2% solely due to such indirect spillover effects. The conclusions remain the same when examining indirect spillover effects on investment within a district-industry combination instead (column 3).

Finally, the coefficient estimates reported in columns 4 to 6 of Table 10 suggest that while the bank resolution had direct effects on firm *i*'s employment decisions as we show in Table 9, it did not lead to any statistically significant spillover effects for firms operating in districts or district-industry pairs with a high share of firms exposed to the shock. The lack of indirect spillover effects for employment and the smaller direct effect in terms of economic magnitude when

compared to investment is consistent with a stronger persistence in employment than in investment decisions.²⁹

4.3.3 Role of firms' internal liquidity. The option for firms to access liquidity from credit lines should be more valuable when internal liquidity is scarce (e.g., Campello et al. 2011). Thus, if the adverse effects of the bank resolution on real outcomes are indeed driven by a reduction in granted credit lines for SMEs, we should observe heterogeneous effects according to their preshock liquidity position—that is, if firms view cash and lines of credit as liquidity substitutes and given the tighter credit line limits, illiquid SMEs might respond to the funding shock by increasing cash holdings while decreasing investment and employment. Berg (2018), for instance, shows in a different setting that while liquid SMEs are able to absorb credit supply shocks by using existing cash buffers, their illiquid counterparts increase cash holdings when a loan application is rejected, cutting noncash assets by more than the requested loan amount, and thus investment and employment.

We analyze this channel explicitly in Table 11 where we split SMEs (panel A) and large firms (panel B) according to their *ex ante* liquidity position—low-liquidity firms (cash holdings-to-assets ratio before the shock lower than the median) versus high-liquidity firms (cash holdings-to-assets ratio before the shock higher than the median). Columns 1, 3, 5, and 7 focus on the full sample of firms with multiple bank relationships, while columns 2, 4, 6, and 8 report the results based on the restricted sample of firms with credit lines granted by at least two different banks to better trace the effects of the supply driven contraction in granted credit lines to real outcomes.

The estimates in columns 3 and 4 show a significant increase in cash holdings by low-liquidity SMEs more exposed to the bail-in. This effect is not present across large firms. Conversely, in line with a precautionary savings motive where firms hold cash as a buffer against adverse shocks (e.g., Duchin, Ozbas, and Sensoy 2010), high-liquidity firms more exposed to the bail-in decreased cash holdings considerably—even though low- and high-liquidity SMEs and large firms were subject to a similar credit supply shock (columns 1 and 2). In economic terms, a one-standard deviation increase in firm exposure to the bail-in leads to an increase in cash holdings for low-liquidity SMEs of at least 11.2%, but a decrease in cash holdings for high-liquidity SMEs of at least 18.8%.

Columns 5 and 6 report the coefficient estimates for investment, while columns 7 and 8 focus on employment. The results show that the negative real effects are concentrated in SMEs with low preperiod levels of internal

²⁹ These results assume symmetric spillover effects for treated and control group firms. Nevertheless, we obtain similar, if not stronger, results when allowing for asymmetric spillover effects for treated and control group firms, as proposed by Berg and Streitz (2019). As in the credit supply shock in Germany analyzed by Huber (2018) and Berg and Streitz (2019), the negative spillover effects on investment are purely driven by the control group firms.

Table 11
Firm exposure to the bail-in and internal liquidity

	$\Delta \log$ $TotalCredit_i$	$\Delta \log$ $CreditLines_i$	$\Delta \log Cash$ $Holdings_i$		$\Delta \log$ $Investment_i$		$\Delta \log No.$ $Employees_i$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	(8)						
<i>Panel A: SMEs</i>							
Firm exposure $\times$ high-liquidity firms	0.003 (0.374)	-0.017*** (-2.653)	-0.188*** (-3.055)	-0.328*** (-4.491)	-0.002 (-0.210)	0.012 (1.091)	-0.002 (-0.420)
Firm exposure $\times$ low-liquidity firms	-0.008 (-1.615)	-0.023*** (-4.341)	0.137*** (2.805)	0.112** (2.611)	-0.028*** (-8.614)	-0.028*** (-5.131)	-0.017*** (-9.258)
No. observations/firms	40,236	13,993	40,236	13,993	40,236	13,993	40,236
No. banks	98	92	98	92	98	92	98
Adj. R^2	.382	.174	.019	.027	.042	.046	.067
<i>Panel B: large firms</i>							
Firm exposure $\times$ high-liquidity firms	-0.020 (-1.315)	-0.014 (-0.109)	-0.214** (-2.443)	-0.686*** (-4.742)	-0.009 (-0.210)	-0.035 (-1.377)	0.030 (1.139)
Firm exposure $\times$ low-liquidity firms	-0.004 (-0.137)	-0.058 (-0.940)	0.009 (0.094)	0.117 (0.602)	-0.034 (-1.359)	-0.015 (-0.577)	-0.002 (-0.078)
No. observations/firms	691	327	691	327	691	327	691
No. banks	16	14	16	14	16	14	16
Adj. R^2	.225	.224	.073	.134	.009	.122	.091
Firm and bank controls	Y	Y	Y	Y	Y	Y	Y
Credit demand	Y	Y	Y	Y	Y	Y	Y
Industry and district FE	Y	Y	Y	Y	Y	Y	Y
No. bank relationships > 1	Y	Y	Y	Y	Y	Y	Y
Credit lines with > 1 banks	N	Y	N	Y	N	Y	N

This table presents estimation results of cross-sectional model 2, where firms are split according to their ex ante liquidity position: low-liquidity (cash holdings-to-total assets ratio before the shock lower than the median) versus high-liquidity firms (cash holdings-to-total assets ratio before the shock higher than the median). The dependent variables are the change in the log level of cash holdings, investment (tangible assets), and employment (number of employees) for each firm between 2013:Q4 and 2015:Q4. Panel A focuses on the subsample of SMEs and panel B on large firms. Firm exposure captures the average exposure of each firm to the bail-in and is computed as the weighted average of bank exposure across all banks lending to a firm, using as weights the preperiod share of total credit from each bank. Bank exposure is the percentage of assets of each bank exposed to the bail-in—that is, the percentage of assets that was effectively bailed-in for the resolved bank, the specific contribution to the *ad hoc* loan to the Resolution Fund granted as part of the resolution for the eight participating banks (as a percentage of assets), and zero otherwise. Bank controls, averaged at the firm level according to the preperiod share of total credit granted to the firm by each bank, are measured as of 2013:Q4 and include bank size (log of total assets), bank ROA (return-on-assets), bank capital ratio (equity to total assets), bank liquidity ratio (liquid to total assets), and bank NPLs (nonperforming loans to total gross loans). Firm controls are also measured before the shock (2013:Q4) and include firm size (log of total assets), firm age (ln(1+age)), firm ROA (net income to total assets), firm capital ratio (equity to total assets), and firm current ratio (current assets to current liabilities). Credit demand is the vector of firm-level dummies estimated in the within-firm regression (column 1 of Table 2). All coefficients are scaled by the corresponding variable's standard deviation, and *t*-statistics (in parentheses) are robust to heteroskedasticity and within-main-bank dependence. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

liquidity, corresponding to those firms that increased cash holdings as a result of the shock. This suggests that while more exposed SMEs with high liquidity before the bail-in were able to use their available internal liquidity buffers to compensate for the binding contraction in granted credit lines and thus maintain employment and investment, low-liquidity SMEs more exposed to the bail-in responded by increasing cash holdings while decreasing investment and employment. These results therefore highlight the “dark side” of precautionary savings first documented by Berg (2018), where an update in beliefs about the optimal level of cash holdings leads to negative real effects. While Berg (2018) compares accepted and rejected loan applicants at a single German bank, we help to generalize his findings by analyzing the entire Portuguese banking sector in the context of supply driven credit line reduction.

A potential concern regarding the results in Table 11 is that low levels of firm liquidity prior to the shock might reflect declining demand for investment given that cash holdings are chosen at least partially based on anticipated investment opportunities (e.g., Cunha and Pollet forthcoming). To help rule out this possibility, in Table IA10 in the Internet Appendix we split high- and low-liquidity SMEs according to the firm-specific preshock asset growth before the shock—that is, below and above the median of the overall sample. Our results hold across the two subsamples. Specifically, SMEs with both lower and higher growth opportunities before the resolution increased cash holdings and decreased investment and employment if they had low levels of internal liquidity, and both decreased cash holdings and maintained employment and investment if they were highly liquid before the shock.

Overall, the results in Tables 8 to 11 show that although there was on average and across the different firm size groups no reduction in overall borrowing after the bank resolution, SMEs still decreased investment and employment. This is explained by smaller firms with low levels of internal liquidity before the shock reacting to the tightening of credit line limits by hoarding cash while at the same time cutting back on investment and employment. The negative impact of the bank resolution shock on investment and employment can thus be explained with heightened liquidity risk.

4.4 Bail-in versus bailout

Our results pointing toward negative real effects after a bank bail-in are particularly relevant given the growing evidence that, even if setting the stage for aggressive risk-taking and future fragility, bank bailouts can be effective in supporting borrowers in the short term.³⁰ Therefore, a final question is whether a bank bailout would generate the same negative effects we show in the paper

³⁰ See, for instance, Giannetti and Simonov (2013) for the Japanese crisis of the 1990s, Augusto and Félix (2014) for the bailouts in Portugal, and Li (2013) for the TARP-funded bailouts in the United States. Laeven and Valencia (2013) examine financial sector interventions in fifty countries after the 2007–2009 financial crisis and also show these improved the value-added growth of financially dependent firms.

for a bail-in. While we cannot make this comparison directly due to the lack of a counterfactual, we shed some light on this issue by exploiting the fact that the bail-in of shareholders and junior bondholders we analyzed so far in this paper differed significantly from the approach taken by the Portuguese authorities during earlier bank failures during the crisis. Notably, in June 2012, three of the largest five banks (CGD, Banco Millennium BCP, and Banco BPI) received government-funded capital injections, as well as the smaller BANIF in December 2012. The bailouts allowed banks to comply with stricter minimum capital requirements defined by the EBA.

To assess the effects of the bailouts on credit supply and real outcomes, we use both within-firm and cross-sectional regressions, with data averaged between the fourth quarter of 2011 and the second quarter of 2012 as the prebailout period and between the fourth quarter of 2012 and the fourth quarter of 2013 as the postbailout period—see Figure IA1 in the Internet Appendix. We have data on 45,062 firms that had relationships with at least two banks, including the four bailed-out banks, for a total of 122,749 firm–bank relationships. Bank exposure is a dummy variable that takes the value of one for bailed-out banks and zero otherwise in the baseline case. Overall, 54% of all firms had a relationship with the bailed-out banks.

The results in Table IA11 in the Internet Appendix show no significant difference in credit growth at the intensive margin between the bailed-out and other banks for the same borrower and no significant variation in firm-level credit supply, investment, or employment with exposure to bailed-out banks, suggesting that the bailouts fulfilled their objectives of protecting borrowers of failing banks.³¹ In summary, we find no evidence of a negative impact of the bank bailouts in 2012 on the relative credit supply by bailed-out versus non-bailed-out banks. Consequently, there was also no relative decline in investment or employment by firms more exposed to the bailed-out banks. While this points to rather sharp differences between bailouts and bail-ins of banks, we urge caution in interpreting this comparison directly since (i) the macroeconomic situation was considerably different during these two episodes; (ii) in contrast with the 2012 bailouts, the bail-in in 2014 that we focus on was accompanied by management changes; and (iii) the public intervention in 2012 was more systemic in nature. Moreover, previous evidence has shown the detrimental impact on bank risk-taking generated by public guarantees such as bailouts (e.g., Dam and Koetter 2012) or even deposit insurance (e.g., Calomiris and Jaremski 2019).³² Instead, despite being an issue outside the scope of this

³¹ In robustness tests, available in Table IA12, we find no significant effects of the bailout for either large firms or SMEs and show that our findings are robust to an alternative measure of bank exposure defined as a continuous treatment variable equal to the injection amount as a share of assets for each of the bailed-out banks and zero otherwise.

³² More generally, government guarantees can distort banks' incentives to raise new equity and the dynamics of bank capital structure over the credit cycle (e.g., Baron forthcoming), as well as exacerbate the negative feedback

paper, bank bail-ins should reduce moral hazard due to creditors' expectation of bearing the losses in case of distress (Schäfer, Schnabel, and Weder 2017).

5. Conclusion

We examine the credit supply and real effects of bank bail-ins by exploiting the unexpected failure and subsequent resolution of a major Portuguese bank. Using matched firm–bank loan-level data, we show that firms more exposed to the bail-in suffered an economically and statistically significant contraction of available credit at the intensive margin. This supply-driven effect is more pronounced among large firms and concentrated on firms whose main lender prior to the shock was the bailed-in bank. While firms more affected by the shock were, on average, able to compensate for the overall credit contraction with funding from other banks they already had relationships with, exposed SMEs were subject to a binding contraction in the quantity of funds available through lines of credit. Those smaller firms with lower preshock internal liquidity responded by increasing precautionary cash holdings and reducing investment and employment.

Our findings show that a well-designed bank resolution framework that includes a bail-in of shareholders and bondholders together with prompt governance and management changes can mitigate the potentially severe negative impact of bank failures on credit supply and thus provide supporting evidence for the move to this new resolution mechanism. While bail-ins can also be a superior alternative to bailouts if the policymakers' objective is to maximize financial sector stability and keep the banks' probability of default low, the trade-off is that such banks will grow more slowly and generate less value after a bail-in, leading to negative effects to the real economy. Thus, our results suggest that it is not enough to recapitalize a bank without resorting to public funds—instead, regulators also need to ensure that the incentives *ex post* are in line with supplying credit to the real economy. In other words, a bank bail-in is not a silver bullet, and only the combination of robust supervisory, regulatory, and resolution frameworks can ensure a sound banking system and minimize the adverse effects of bank distress on the real economy.

References

- Abowd, J. M., F. Kramarz, and D. N. Margolis. 1999. High wage workers and high wage firms. *Econometrica* 67(2):251–333.
- Aiyar, S., C. W. Calomiris, and T. Wieladek. 2014. Does macro-prudential regulation leak? Evidence from a UK policy experiment. *Journal of Money, Credit and Banking* 46(s1):181–214.

loop between banks and sovereigns that characterized the euro area crisis (e.g., Crosignani, Faria-e Castro, and Fonseca 2020). As highlighted by Granja, Matvos, and Seru (2017) following the FDIC experience in dealing with failing banks in the United States, the risk-taking incentives created when not failing a bank should nonetheless be weighted against those of misallocating a bank's assets when selling failed banks.

- Almeida, H., M. Campello, and M. S. Weisbach. 2004. The cash flow sensitivity of cash. *Journal of Finance* 59(4):1777–1804.
- Ashcraft, A. B. 2005. Are banks really special? New evidence from the FDIC-induced failure of healthy banks. *American Economic Review* 95(5):1712–1730.
- Augusto, F., and S. Félix. 2014. The impact of bank recapitalization on firms' access to credit: evidence from Portugal. *Banco de Portugal Financial Stability Papers* 2-2014.
- Avgouleas, E., and C. Goodhart. 2015. Critical reflections on bank bail-ins. *Journal of Financial Regulation* 1(1):3–29.
- Banco de Portugal. 2014a. Address by Mr Carlos da Silva Costa, Governor of Banco de Portugal, on the decision to apply a resolution measure to Banco Espírito Santo. URL: <https://www.bportugal.pt/en/intervencoes/address-governor-banco-de-portugal-carlos-da-silva-costa-decision-regarding-application>.
- . 2014b. Deliberation of the Board of Directors of Banco de Portugal of 3 August 2014. URL: https://www.bportugal.pt/sites/default/files/deliberation_3_aug_2014_8pm.pdf
- Baron, M. Forthcoming. Countercyclical bank equity issuance. *Review of Financial Studies*.
- Beck, T. 2011. Bank resolution: a conceptual framework. *Financial Regulation at the Crossroads: Implications for Supervision, Institutional Design and Trade* 12, 53.
- Beck, T., H. Degryse, R. De Haas, and N. Van Horen. 2018. When arm's length is too far: Relationship banking over the credit cycle. *Journal of Financial Economics* 127(1):174–196.
- Beck, T., D. Radev, I. Schnabel. 2020. Bank resolution regimes and systemic risk. CEPR Discussion Paper No. DP14724.
- Berg, T. 2015. Playing the devil's advocate: The causal effect of risk management on loan quality. *Review of Financial Studies* 28(12):3367–3406.
- . 2018. Got rejected? Real effects of not getting a loan. *Review of Financial Studies* 31(12):4912–4957.
- Berg, T., A. Saunders, S. Steffen, and D. Streitz. 2017. Mind the gap: The difference between US and European loan rates. *Review of Financial Studies* 30(3):948–987.
- Berg, T., and D. Streitz. 2019. Handling spillover effects in empirical research. Working Paper, Frankfurt School of Finance & Management.
- Berger, A., C. P. Himmelberg, R. A. Roman, and S. Tsyplakov. 2020. Bank bailouts, bail-ins, or no regulatory intervention? A dynamic model and empirical tests of optimal regulation. Working Paper, University of South Carolina.
- Bernanke, B. S. 1983. Nonmonetary effects of the financial crisis in the propagation of the Great Depression. *American Economic Review* 73(3):257–276.
- Bernard, B., A. Capponi, and J. E. Stiglitz. 2019. Bail-ins and bail-outs: Incentives, connectivity, and systemic stability. Working Paper, Columbia University.
- Berton, F., S. Mocetti, A. F. Presbitero, and M. Richiardi. 2018. Banks, firms, and jobs. *Review of Financial Studies* 31(6):2113–2156.
- Bertrand, M., E. Dufo, and S. Mullainathan. 2004. How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics* 119(1):249–275.
- BES. 2014. Balanco Individual e Notas Explicativas a 4 de Agosto de 2014 (in Portuguese). URL: <http://www.bes.pt/Comunicados/20150807%20FR56737.pdf>.
- Bolton, P., X. Freixas, L. Gambacorta, and P. E. Mistrulli. 2016. Relationship and transaction lending in a crisis. *Review of Financial Studies* 29(10):2643–2676.
- Bonaccorsi di Patti, E., and E. Sette. 2016. Did the securitization market freeze affect bank lending during the financial crisis? Evidence from a credit register. *Journal of Financial Intermediation* 25:54–76.

- Bonfim, D., G. Cerqueiro, H. Degryse, and S. Ongena. 2020. On-site inspecting zombie lending. Working Paper, Banco de Portugal.
- Bottero, M., S. Lenzu, and F. Mezzanotti. 2020. Sovereign debt exposure and the bank lending channel: Impact on credit supply and the real economy. *Journal of International Economics* 126:103328.
- Brown, M., I. Evangelou, and H. Stix. 2018. Banking crises, bail-ins and money holdings. Working Paper, Central Bank of Cyprus.
- Calomiris, C. W., and M. Jaremski. 2019. Stealing deposits: Deposit insurance, risk-taking, and the removal of market discipline in early 20th-century banks. *Journal of Finance* 74(2):711–754.
- Calomiris, C. W., and J. R. Mason. 2003. Consequences of bank distress during the Great Depression. *American Economic Review* 93(3):937–947.
- Cameron, A. C., J. B. Gelbach, and D. L. Miller. 2008. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90(3):414–427.
- Campello, M., E. Giambona, J. R. Graham, and C. R. Harvey. 2011. Liquidity management and corporate investment during a financial crisis. *Review of Financial Studies* 24(6):1944–1979.
- Carvalho, D., M. A. Ferreira, and P. Matos. 2015. Lending relationships and the effect of bank distress: Evidence from the 2007–2009 financial crisis. *Journal of Financial and Quantitative Analysis* 50(06):1165–1197.
- Chodorow-Reich, G. 2014. The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis. *Quarterly Journal of Economics* 129(1):1–59.
- Cingano, F., F. Manaresi, and E. Sette. 2016. Does credit crunch investment down? New evidence on the real effects of the bank-lending channel. *Review of Financial Studies* 29(10):2737–2773.
- Colliard, J.-E., and D. Gromb. 2019. Financial restructuring and resolution of banks. Working Paper, HEC Paris.
- Conlon, T., and J. Cotter. 2014. Anatomy of a bail-in. *Journal of Financial Stability* 15:257–263.
- Crosignani, M., M. Faria-e Castro, and L. Fonseca. 2020. The (unintended?) consequences of the largest liquidity injection ever. *Journal of Monetary Economics* 112:97–112.
- Cunha, I., and J. Pollet. Forthcoming. Why do firms hold cash? Evidence from demographic demand shifts. *Review of Financial Studies*.
- Dam, L., and M. Koetter. 2012. Bank bailouts and moral hazard: Evidence from Germany. *Review of Financial Studies* 25(8):2343–2380.
- Degryse, H., O. De Jonghe, S. Jakovljević, K. Mulier, and G. Schepens. 2019. Identifying credit supply shocks with bank-firm data: Methods and applications. *Journal of Financial Intermediation* 40:100813.
- Dell'Ariccia, G., M. S. M. Peria, D. O. Igan, E. A. Awadzi, M. Dobler, and D. Sandri. 2018. Trade-offs in bank resolution. Staff Discussion Notes No. 18/02, International Monetary Fund.
- Dewatripont, M. 2014. European banking: Bailout, bail-in and state aid control. *International Journal of Industrial Organization* 34:37–43.
- Duchin, R., O. Ozbas, and B. A. Sensoy. 2010. Costly external finance, corporate investment, and the subprime mortgage credit crisis. *Journal of Financial Economics* 97(3):418–435.
- Economist. 2014. Banco Espírito Santo: Sharing the pain. August 9, 2014. URL: <https://www.economist.com/finance-and-economics/2014/08/09/sharing-the-pain>.
- FT. 2014. Banco Espírito Santo: Family fortunes. September 11, 2014. URL: <https://www.ft.com/content/a63a4a56-32c0-11e4-93c6-00144feabdc0>.
- Garcia, A. R. 2016. Banco Espírito Santo, SA: Resolution via a bridge bank including a re-transfer. In: *Bank resolution and "bail-in" in the EU: Selected case studies pre and post BRRD*. Washington, DC: World Bank:52–60.
- Giannetti, M., and A. Simonov. 2013. On the real effects of bank bailouts: Micro evidence from Japan. *American Economic Journal: Macroeconomics* 5(1):135–167.

- Granja, J., G. Matvos, and A. Seru. 2017. Selling failed banks. *Journal of Finance* 72(4):1723–1784.
- Gropp, R., H. Hakenes, and I. Schnabel. 2011. Competition, risk-shifting, and public bail-out policies. *Review of Financial Studies* 24(6):2084–2120.
- Gropp, R., T. Mosk, S. Ongena, and C. Wix. 2019. Banks response to higher capital requirements: Evidence from a quasi-natural experiment. *Review of Financial Studies* 32(1):266–299.
- Havemann, R. 2019. Can creditor bail-in trigger contagion? The experience of an emerging market. *Review of Finance* 23(6):1155–1180.
- Huber, K. 2018. Disentangling the effects of a banking crisis: Evidence from German firms and counties. *American Economic Review* 108(3):868–98.
- Hüser, A.-C., G. Hałaj, C. Kok, C. Perales, and A. van der Kraaij. 2018. The systemic implications of bail-in: A multi-layered network approach. *Journal of Financial Stability* 38:81–97.
- Ippolito, F., J.-L. Peydró, A. Polo, and E. Sette. 2016. Double bank runs and liquidity risk management. *Journal of Financial Economics* 122(1):135–154.
- Iyer, R., J.-L. Peydró, S. Da-Rocha-Lopes, and A. Schoar. 2014. Interbank liquidity crunch and the firm credit crunch: Evidence from the 2007–2009 crisis. *Review of Financial Studies* 27(1):347–372.
- Jiménez, G., J. A. Lopez, and J. Saurina. 2009. Empirical analysis of corporate credit lines. *Review of Financial Studies* 22(12):5069–5098.
- Jiménez, G., A. Mian, J.-L. Peydró, and J. Saurina. Forthcoming. The real effects of the bank lending channel. *Journal of Monetary Economics*.
- Jiménez, G., S. Ongena, J.-L. Peydró, and J. Saurina. 2017. Macroprudential policy, countercyclical bank capital buffers, and credit supply: Evidence from the Spanish dynamic provisioning experiments. *Journal of Political Economy* 125(6):2126–2177.
- Kahn, C., and A. Winton. 2004. Moral hazard and optimal subsidiary structure for financial institutions. *Journal of Finance* 59(6):2531–2575.
- Keister, T., and Y. Mitkov. 2019. Bailouts, bail-ins and banking crises. Working Paper, University of Bonn.
- Khwaja, A. I., and A. Mian. 2008. Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review* 98(4):1413–1442.
- Klimek, P., S. Poledna, J. D. Farmer, and S. Thurner. 2015. To bail-out or to bail-in? Answers from an agent-based model. *Journal of Economic Dynamics and Control* 50:144–154.
- Laeven, L., and F. Valencia. 2013. The real effects of financial sector interventions during crises. *Journal of Money, Credit and Banking* 45(1):147–177.
- Lambrecht, B. M., and A. S. Tse. 2019. Liquidation, bailout, and bail-in: Insolvency resolution mechanisms and bank lending. CEPR Discussion Paper No. DP13734.
- Li, L. 2013. TARP funds distribution and bank loan supply. *Journal of Banking & Finance* 37(12):4777–4792.
- Novo Banco. 2014. *2014 annual report*.
- . 2015. *2015 annual report*.
- Paravisini, D., V. Rappoport, and P. Schnabl. 2020. Specialization in bank lending: Evidence from exporting firms. CEPR Discussion Paper No. 12156.
- Peek, J., and E. S. Rosengren. 2000. Collateral damage: Effects of the Japanese bank crisis on real activity in the United States. *American Economic Review* 90(1):30–45.
- Philippon, T., and A. Salord. 2017. Bail-ins and bank resolution in Europe: a progress report. *Geneva Reports on the World Economy Special Report 4*.

- Puri, M., J. Rocholl, and S. Steffen. 2011. Global retail lending in the aftermath of the US financial crisis: Distinguishing between supply and demand effects. *Journal of Financial Economics* 100(3):556–578.
- Reis, R. 2015. Looking for a success in the euro crisis adjustment programs: The case of Portugal. *Brookings Papers on Economic Activity* 2015(2):433–458.
- Santos, J. A. 2011. Bank corporate loan pricing following the subprime crisis. *Review of Financial Studies* 24(6):1916–1943.
- Schäfer, A., I. Schnabel, and B. Weder. 2017. Expecting bail-in? Evidence from European Banks. Working Paper, University of Bonn.
- Segura, A., and S. Vicente. 2019. Bank resolution and public backstop in an asymmetric banking union. Working Paper, Bank of Italy.
- Silva, A. F. 2019. Strategic liquidity mismatch and financial sector stability. *Review of Financial Studies* 32(12):4696–4733.
- Sufi, A. 2009. Bank lines of credit in corporate finance: An empirical analysis. *Review of Financial Studies* 22(3):1057–1088.
- Walther, A., and L. White. Forthcoming. Rules versus discretion in bank resolution. *Review of Financial Studies*.